

BIOLOGY GRADE 10: CELL BIOLOGY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 1.0: Cell Biology and Biodiversity
Sub-Strand	Sub-Strand 1.3: Cell Biology — Cell Structure and Specialisation
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Cell as the basic unit of life — historical background (Hooke, Schleiden, Schwann, Virchow) – Types of cells: prokaryotic (bacteria such as <i>E. coli</i>) vs. eukaryotic (plant and animal cells) – Ultra-structure of the animal cell: cell membrane, nucleus, mitochondria, ribosomes, endoplasmic reticulum, Golgi apparatus, lysosomes, centrioles, cytoplasm – Ultra-structure of the plant cell: cell wall (cellulose), chloroplasts, large central vacuole, plasmodesmata — and how these differ from animal cells – Cell organelles and their functions — detailed functional roles of each organelle – Cell specialisation: how cells differentiate to perform specific functions (e.g., red blood cells, guard cells, root hair cells, muscle cells, nerve cells) – Tissues, organs and organ systems — hierarchy from cell → tissue → organ → organ system → organism – Comparison of plant and animal cells — similarities and differences in structure and function
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe the ultra-structure of plant and animal cells using diagrams and models, b) identify and state the functions of cell organelles in plant and animal cells, c) distinguish between prokaryotic and eukaryotic cells with reference to their structural features, d) explain how cell specialisation enables cells to perform specific functions in living organisms, e) construct a hierarchy from cell to organism, relating cell structure to tissue, organ and organ system organisation, f) appreciate the relationship between cell structure and function in maintaining life processes.
Core Competencies	– Communication and Collaboration: Learners work in small groups to construct annotated 3D cell models using locally available materials (bottle tops, avocado seeds, maize grains, plastic bags) and present their models to the class, justifying their choices of materials for each organelle. – Critical Thinking and Problem Solving: Learners analyse electron micrograph images of different specialised cells (e.g., a Kenyan blood smear showing red blood cells, a maize leaf cross-section) and reason about why each cell's structure suits its function. – Imagination and Creativity: Learners design and build physical or digital models of plant and animal cells, making creative decisions about how to represent organelles using everyday Kenyan materials found at home or in the market. – Self-efficacy: Learners reflect on how understanding cell specialisation connects to real career pathways (medicine,

	agriculture, biotechnology) relevant to Kenya's development goals under Vision 2030.
Core Values	– Responsibility: The learner responsibly handles microscopes and prepared slides of onion epidermis, cheek cells, or other specimens, ensuring equipment is cleaned and stored correctly after use. – Respect: The learner appreciates diverse contributions from peers during group model-building activities and discussions about cell structure and function. – Unity: The learner collaborates equitably in small groups during practical activities, ensuring all members participate in constructing and presenting cell models. – Social Justice: The learner discusses how access to medical technologies rooted in cell biology (e.g., malaria diagnosis via blood smear microscopy — relevant in Kisumu and the Lake Victoria region) affects communities across Kenya.
Science & Engineering Practices	– SEP 1 — Asking Questions and Defining Problems: Learners generate and record investigable questions on the Driving Question Board (DQB) about why cells look different, e.g., "Why does a red blood cell have no nucleus but a cheek cell does?" – SEP 2 — Developing and Using Models: Learners build annotated 3D models of plant and animal cells using locally available materials and revise them as understanding deepens across the 8-lesson sequence. – SEP 4 — Analysing and Interpreting Data: Learners examine photomicrographs and electron micrographs of specialised cells (e.g., guard cells from a kale/sukuma wiki leaf, root hair cells from a bean plant) and identify structural adaptations. – SEP 6 — Constructing Explanations: Learners write and discuss evidence-based explanations for how a cell's structure (e.g., the large surface area of a root hair cell) relates to its function (e.g., water and mineral absorption). – SEP 8 — Obtaining, Evaluating and Communicating Information: Learners research Kenyan scientists in cell biology and biotechnology (e.g., researchers at KEMRI and JKUAT), evaluate sources for credibility, and present findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Health Education: Learners connect cell biology to understanding diseases prevalent in Kenya — e.g., how the malaria parasite (Plasmodium) invades and destroys red blood cells, linking cell structure to public health in endemic counties such as Kisumu and Homa Bay. – Environmental Conservation: Learners discuss how plant cell structures (chloroplasts, cell walls) are fundamental to photosynthesis and food production, reinforcing the importance of conserving vegetation in Kenya's agroecological zones such as the Rift Valley and Central highlands. – Safety and Security: Learners observe safe laboratory practices when preparing wet mount slides (e.g., of onion epidermal cells or cheek cells), handling glass coverslips, and using microscopes responsibly. – Life Skills Education (Critical Thinking): Learners critically evaluate claims about cell-based products (e.g., stem cell therapies advertised in Kenyan media) using their knowledge of cell structure and specialisation. – Career Guidance: Learners relate cell biology knowledge to careers in medicine, nursing, biotechnology, agriculture, and food science within the Kenyan job market and Kenya's Big Four Agenda priorities.
Career Connections	– Medical Doctor / Clinical Officer: Understanding cell structure and specialisation underpins diagnosis and treatment of cellular diseases such as sickle cell anaemia (prevalent in western Kenya) and cancer. – Medical Laboratory Technologist: Directly applies knowledge of cell types and microscopy (as practised at facilities like Kenyatta National Hospital) to diagnose diseases from blood smears and tissue samples. – Agricultural Scientist / Agronomist: Uses knowledge of plant cell specialisation and tissue organisation to improve crop yields of staple foods like maize, beans, and sorghum grown across Kenya. – Biotechnologist / Genetic Engineer: Applies cell biology principles at institutions such as KEMRI and JKUAT to develop GMO crops, vaccines, and diagnostic tools relevant to Kenya's health and food security goals. – Nurse / Midwife: Applies understanding of human cell organisation and organ systems in patient care across Kenya's county hospitals and health centres. – Food Scientist / Nutritionist: Relates cell and tissue biology to the nutritional quality and processing of Kenyan foods such as

	milk, meat, and fermented products like mursik.
Focus for Lessons	This 8-lesson unit anchors learning in the puzzling observation that all living organisms — from a single-celled amoeba in a Nairobi pond to a maize plant in the Rift Valley to a Kenyan marathon runner — are built from the same fundamental unit: the cell, yet cells look radically different from one another. Learners move from examining familiar organisms to building detailed understanding of cell ultra-structure, organelle function, and specialisation, using models, microscopy, and data analysis. The unit consistently connects cell biology to Kenya-relevant health, agricultural, and biotechnology contexts, ensuring learning is grounded in students' lived experiences.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "Why do the cells in my body — and in the sukuma wiki on my plate — look so different from each other, and how does each cell's structure allow it to do its specific job?" KICD KEY INQUIRY QUESTIONS: 1. What are the structural differences between plant and animal cells? 2. How does cell specialisation enable different cells to perform specific functions in living organisms?
Anchoring Phenomenon	ANCHORING PHENOMENON: A Form 4 student from Kisumu donates blood at the local county hospital. The medical laboratory technologist prepares a blood smear, stains it, and examines it under a microscope. The student is astonished to see that the red blood cells look like flattened biconcave discs with NO nucleus — yet the white blood cells right next to them have large, visible nuclei. Stranger still, when the same student chews a piece of sukuma wiki (kale) from their family's shamba and a teacher prepares a microscope slide of the leaf, the cells are rectangular, have thick green structures inside, and appear to have a large water-filled sac pushing everything to the side. HOW CAN ALL THESE STRUCTURES — a human blood cell, a white blood cell, a leaf cell — all be called "cells" yet look and behave so completely differently? What do they share? What is different, and WHY? This puzzle drives the entire unit: students must build a model of what a "generic" cell looks like, then explain how and why specialised cells deviate from that model to perform specific jobs in the body and in plants.
Storyline Thread	Lesson 1 — PREDICT: "What is a cell?" Students observe the anchoring phenomenon (blood smear image + sukuma wiki leaf image side by side). Without prior instruction, students sketch what they think a "basic cell" looks like and predict: what structures must ALL cells share? Predictions posted to the Driving Question Board (DQB). Teacher introduces the historical thread: Hooke, Schleiden, Schwann, Virchow — why did it take so long for humans to discover cells? Lesson 2 — OBSERVE (Microscopy Practical 1): "Seeing cells for the first time" Students prepare wet mount slides of onion epidermal cells AND their own cheek cells (cotton bud method, safely). They observe, draw, and label what they see under low and high power. Students record DIFFERENCES between the two cell types. DQB updated: new questions added about structures they cannot yet explain (e.g., "What is the green stuff in the plant cell?"). Lesson 3 — EXPLAIN (Organelle Structure & Function — Animal Cell): "What is inside the cell doing all the work?" Using electron micrograph images and annotated diagrams, learners systematically identify and learn the function of each animal cell organelle — nucleus, mitochondria, ribosomes, ER, Golgi, lysosomes, centrioles. Students link each organelle to a real-life factory analogy set in a Nairobi industrial park (nucleus = managing director's office; mitochondria = generator room; etc.). Students return to DQB and answer early questions. Lesson 4 — EXPLAIN (Organelle Structure & Function — Plant Cell): "How is a plant cell like and unlike an animal cell?" Focus on plant-specific organelles: chloroplasts (photosynthesis in sukuma wiki leaves), cell wall (rigidity and wilting phenomenon), central vacuole (turgor pressure and the wilting bean supporting phenomenon). Students complete a structured comparison table. Model-building begins: students start constructing labelled 2D diagrams of both cell types.

	Lesson 5 — OBSERVE + EXPLAIN (Prokaryote vs. Eukaryote): "Are bacteria cells the same as our cells?" Revisit the fermenting uji supporting phenomenon. Students examine images/diagrams of bacterial cells (E. coli) vs. human cells. Identify key differences: no membrane-bound nucleus, no membrane-bound organelles, smaller size, presence of cell wall in bacteria. Students update their DQB and revise their models to add a "prokaryote" column. Lesson 6 — EXPLAIN (Cell Specialisation): "Why don't all cells look the same?" Students investigate 6 specialised cells: red blood cells (no nucleus → more haemoglobin space), guard cells (bean-shaped → control stomatal opening in sukuma wiki), root hair cells (elongated → large surface area in bean seedling), muscle cells (elongated with many mitochondria → contraction), nerve cells (long axons → signal transmission), and sperm cells (flagellum → motility). Students match structural features to functions using evidence cards. Sickle cell supporting phenomenon is revisited: students explain why the abnormal shape disrupts function. Lesson 7 — MODEL BUILDING + DQB Revision: "Building our complete cell model" Students construct annotated 3D models of a plant OR animal cell using locally available materials (avocado seed = nucleus, plastic bag = cell membrane, bottle tops = mitochondria, green paper = chloroplasts, maize grain = ribosomes, etc.). Groups present models, explain organelle choices, and peer-assess. DQB is formally revisited: which questions have now been answered? Which remain? Students record remaining questions as "bridges" to future sub-strands (Chemicals of Life, Plant Physiology). Lesson 8 — DRIVING QUESTION SYNTHESIS + Assessment: "So why DO cells look so different?" Students return to the original anchoring phenomenon (blood smear + sukuma wiki leaf). Each student writes a full evidence-based explanation answering the driving question, referencing: (a) shared organelles all cells have, (b) organelles unique to plant or animal cells, (c) how specialisation removes or modifies organelles for a specific function, and (d) one Kenya-relevant example (sickle cell, malaria parasite, root hair absorption). Class constructs a final consensus model on the board. Teacher previews Sub-Strand 1.4 (Chemicals of Life): "Now that we know what cells look like — what are they actually MADE of?"
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — The Fermenting Uji: A family in Kisii leaves a pot of uji (fermented porridge) uncovered overnight. By morning it smells sour and has bubbles. Under a microscope, tiny single-celled organisms (yeast and bacteria) are visible moving and dividing. Students ask: are these the same kind of "cell" as in our bodies? (Introduces prokaryote vs. eukaryote distinction.) – SUPPORTING PHENOMENON 2 — The Wilting Bean Plant: A bean plant (common across Kenyan smallholder farms) is left without water for two days and wilts dramatically. When watered, it recovers within hours. Students observe that plant cells have a large central vacuole that fills with water — what role does this organelle play, and why don't animal cells behave the same way? – SUPPORTING PHENOMENON 3 — The Sickle Cell Crisis: A student reads a news report about a child from western Kenya being admitted to hospital during a "sickle cell crisis." Photographs show that the child's red blood cells are sickle-shaped instead of biconcave disc-shaped and are blocking small blood vessels. Students ask: how does a change in cell shape — at the microscopic level — cause so much pain and organ damage? – SUPPORTING PHENOMENON 4 — The Root Hair Close-Up: A freshly pulled bean seedling from the school garden shows tiny, hair-like projections on the roots visible to the naked eye. Under a microscope these are revealed to be single elongated cells. Students ask: why are these cells so long and thin, and how does their shape help the plant absorb water from Kenyan red soil?

LESSON 1 (80 minutes (double lesson)): What Is a Cell? Launching the Phenomenon — Blood, Sukuma Wiki, and the Mystery of Life's Building Blocks

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon and elicit students' prior ideas about cell structure, establishing the investigative question that will drive the entire unit.
Knowledge	By the end of this lesson, the learner should be able to: (a) state that all living organisms are made of cells; (b) identify that cells can vary dramatically in shape, size, and internal structures while still being classified as cells; (c) recall key milestones in the historical development of the cell theory (Hooke, Schleiden, Schwann, Virchow).
Skills	By the end of this lesson, the learner should be able to: (a) sketch and annotate a predicted 'basic cell' diagram based on prior knowledge and observation; (b) formulate testable questions arising from the anchoring phenomenon; (c) record observations from microscope images systematically using a structured observation table.
Attitudes	The learner appreciates that scientific understanding is built incrementally over time through the curiosity and collaborative work of many scientists, and values asking questions as the starting point of inquiry.
Key Inquiry Question	1. What are the structural differences between plant and animal cells? 2. How does cell specialisation enable different cells to perform specific functions in living organisms?
Purpose in Storyline	This is the OPENING lesson of the unit. Its sole purpose is to create productive confusion — students encounter the anchoring phenomenon (blood smear vs. sukuma wiki leaf cell images) and realise they cannot yet explain why cells of the same organism, or across organisms, look so radically different. All predictions and questions generated here are posted to the Driving Question Board (DQB) and will be revisited and revised across all 6 lessons of the sub-strand. No definitive answers are given in this lesson; the goal is to surface prior knowledge, generate curiosity, and establish the investigative frame.
Safety Notes	If physical microscopes and prepared slides are used during the Observe phase: (1) Handle glass slides with care — report any breakage to the teacher immediately; do not attempt to pick up glass shards. (2) Do not touch the microscope lens with fingers. (3) If staining reagents (e.g., iodine) are available, avoid contact with skin and eyes; wash hands after use. (4) No safety concerns arise from viewing projected digital images. Teacher should establish lab safety norms at the start of the lesson if conducted in the laboratory.

B. LESSON OVERVIEW

Students are confronted — without prior instruction — with two side-by-side microscope images: (1) a stained human blood smear showing biconcave, anucleate red blood cells alongside nucleated white blood cells, and (2) a microscope image of a sukuma wiki (kale) leaf cell showing rectangular shape, chloroplasts, a large central vacuole, and a cell wall. The lesson opens with a silent observation, followed by individual prediction sketches of what a 'basic cell' looks like. Student predictions are shared, debated, and posted to the newly created Driving Question Board (DQB). The teacher then introduces the historical thread — Hooke, Schleiden, Schwann, and Virchow — to show that even professional scientists took centuries to agree on what a cell is. The lesson closes with students writing their best current question about cells on a sticky note for the DQB. No formal definitions or diagrams are provided by the teacher; understanding is built through student sensemaking of the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Students enter to find two large projected images displayed silently on the screen — no labels, no title. Image A shows a stained blood smear (red blood cells and white blood cells visible). Image B shows a sukuma wiki leaf cell under a microscope (chloroplasts, large central vacuole, rectangular shape, cell wall visible). Students are given 3 minutes of silent, individual 'Notice and Wonder' time. Each student writes in their exercise book: at least 3 things they NOTICE and at least 2 things they WONDER. After silent time, students share with a shoulder partner (2 min). Each student then independently sketches — in pencil, in their exercise books — what they believe a 'basic cell' (a cell that could represent ALL cells) looks like, and labels any structures they think must be present. Beneath their sketch, they write: 'I predict all cells must have: ____' and 'I predict cells can be different because: ____'. Predictions are NOT corrected by the teacher at this stage.	As students enter, teacher stands silently at the front, gesturing to the projected images — do NOT introduce or explain. After 3 minutes, say: 'Without talking to anyone, write in your book — three things you notice, two things you wonder. You have 3 minutes. Go.' [START 3-MIN TIMER VISIBLY ON BOARD.] Circulate silently; do not answer questions. After sharing time, say: 'Now I want you to do something bold. Without me teaching you anything — draw what YOU think a basic cell looks like. Label everything you believe is inside. You have 5 minutes.' [START 5-MIN TIMER.] Cold-call 4 students to share their sketches by holding books up or drawing on the board. Use these exact prompts — to student 1: 'Talk us through your drawing — what did you include and WHY?' [WAIT 10 SECONDS after the question before accepting response.] To student 2: 'What is ONE thing your drawing has that [Student 1]'s does not?' [WAIT 10 SECONDS.] To student 3: 'Look at Image A on the board — the blood cells. Do those cells match YOUR drawing? What is different?' [WAIT 15 SECONDS.] To student 4: 'Look at Image B — the sukuma wiki cell. What does that cell have that surprises you most?' [WAIT 15 SECONDS.] Teacher records ALL proposed cell structures on the board in two columns: 'We agree all cells have:' 'We are NOT sure about:'. Do NOT resolve disagreements — preserve	NOTICE AND WONDER + PREDICT-BEFORE-INSTRUCTION: Students activate prior knowledge and generate personal hypotheses before encountering any new information. This strategy surfaces misconceptions (e.g., 'all cells are round', 'all cells have a nucleus') that the teacher can deliberately address in later lessons. By making predictions public, students become personally invested in testing and revising them — the core of CBE inquiry. The two-column class record externalises the class's collective prior knowledge and creates a visible 'gap' that motivates learning.	OBSERVABLE INDICATOR: Every student has a labelled cell sketch in their exercise book with at least one written prediction sentence. Teacher circulates and looks for: (1) presence or absence of nucleus in student sketches — a key diagnostic for later misconception resolution; (2) whether students distinguish between plant and animal cell features in Image A vs. B; (3) quality of 'wonder' questions — surface-level ('why is it green?') vs. structural ('why does one cell have a thick border and the other doesn't?'). Teacher mentally notes 2–3 students with particularly generative questions to cold-call during the DQB phase. No marks awarded; this is diagnostic only.

		productive tension.		
Observe Phase  VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Students receive a structured OBSERVATION TABLE (printed or copied from the board) with three columns: 'Feature Observed', 'Blood Cell (Red)', 'Blood Cell (White)', 'Sukuma Wiki Leaf Cell'. Teacher projects each image in turn at high resolution. Students complete the table systematically — describing shape, colour after staining, visible internal structures, and approximate relative size. After completing their own tables individually (8 minutes), students compare with a group of 4 and reach a consensus table on one shared piece of paper. Groups are then directed to circle THREE features that appear in ALL three cell types — these are candidate 'universal cell features'. Groups also star ONE feature that appears in ONLY ONE of the three cell types — their biggest anomaly. A class gallery-walk is simulated: each group reads out their three circled 'universal' features. Teacher tallies on the board which features are most commonly agreed upon across groups. Students then read a short 1-page historical narrative (provided on paper or screen) titled: 'The Long Road to the Cell: From Hooke's Cork to Virchow's Omnis cellula e cellula' — covering Robert Hooke (1665, cork cells, Micrographia), Matthias Schleiden (1838, plant cells), Theodor Schwann (1839, animal cells), Rudolf Virchow (1855, cells come from cells). Students annotate the reading by underlining: in blue — 'what the scientist observed', in red — 'what conclusion they drew'.	Project Image A (blood smear) first. Say: 'You have 4 minutes to fill in the first two columns of your observation table — describe exactly what you SEE, not what you think you know. Be a detective, not a textbook.' [START 4-MIN TIMER.] Then project Image B (sukuma wiki). Say: 'Now 4 minutes for the sukuma wiki leaf cell column.' [START 4-MIN TIMER.] During group comparison: 'Your group has 3 minutes. Find THREE features you all agree appear in every single cell on this board. Find ONE feature that is strangest — that only one type of cell seems to have.' [CIRCULATE — listen for the word 'nucleus'; if a group says 'the red blood cell has no nucleus', crouch to their level and say: 'That is a fascinating observation. Write that down exactly. We are going to chase that question all unit long.'] During the historical reading: 'As you read, I want you to be thinking about one question: WHY did it take from 1665 to 1855 — almost 200 years — for scientists to agree that ALL living things are made of cells? What was making it so hard?' After reading, cold-call 3 students: Student 1 — 'What did Robert Hooke actually see when he looked at cork?' [WAIT 10 SECONDS.] Student 2 — 'Schwann worked with animal tissue. Why do you think it was harder to see animal cells than plant cells?' [WAIT 15 SECONDS — this is a higher-order question; accept all reasonable answers.] Student 3 — 'Virchow said Omnis cellula e cellula — every cell from a cell. How does that idea connect	STRUCTURED OBSERVATION TABLE + ANNOTATED READING: The observation table forces students to move from vague impressions to precise, feature-level descriptions — a core NGSS Science and Engineering Practice (SEP 3: Planning and Carrying Out Investigations; SEP 4: Analysing and Interpreting Data). The annotated reading (observation vs. conclusion) models the distinction between empirical evidence and scientific inference — a critical epistemological habit. The historical narrative contextualises the difficulty of the problem: if brilliant scientists took 200 years, students should expect the unit's central question to be genuinely hard and worth pursuing.	OBSERVABLE INDICATOR: Completed observation tables with entries in all columns. Teacher looks for: (1) Whether students record 'no nucleus' for red blood cells — this is the highest-value observation and will be the anchor for the specialisation lessons. (2) Whether students use descriptive language ('irregular, lobed nucleus' for white blood cells) vs. vague language ('has stuff inside'). (3) During cold-call on Virchow: can students connect 'cells from cells' to the idea that blood cells were once produced by dividing stem cells — a connection that previews specialisation? Award a visible group tally mark on the board to the first group that correctly identifies 'cell membrane' as a universal feature — this gamifies the observation without grading it.

		to the blood smear image we opened class with?' [WAIT 15 SECONDS.]		
Explain Phase  VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Teacher leads a whole-class 'Claim — Evidence — Reasoning' (CER) activity. Using only the evidence gathered in Phase 2 (observation tables and historical reading — NO new information introduced), each student writes a CER response in their exercise book to the prompt: 'CLAIM: Based on our observations today, I think all cells must have at least ____ [name the structures]. EVIDENCE: I know this because in the images we observed, I could see ____.' REASONING: This matters because if a structure is in every cell type we looked at, then ____.' After writing individually (5 minutes), teacher cold-calls 3 students to read their claims aloud. Class votes (thumbs up / thumbs sideways / thumbs down) on whether each claim is supported by the evidence shown today. Teacher scribes the most evidence-supported claim on a large sheet of paper labelled 'OUR BEST CURRENT EXPLANATION — Lesson 1'. This is visibly posted on the classroom wall. Teacher then introduces the formal driving question of the unit verbally (does NOT answer it): 'Why do the cells in my body — and in the sukuma wiki on my plate — look so different from each other, and how does each cell's structure allow it to do its specific job?' Students write this question at the top of a new page in their books — this is their personal unit driving question page.	Say: 'We have not read a textbook today. We have not copied a definition. But you have been doing real science — observing, noticing, wondering, reading primary sources. Now I want you to make a CLAIM based only on what WE found in this room today. You have 5 minutes to write your CER. Go.' [START 5-MIN TIMER. CIRCULATE — if a student writes 'all cells have a nucleus', do not correct; instead ask: 'Are you sure? Look back at your observation table for the red blood cell column.' If a student writes 'all cells have a cell membrane', say quietly: 'That is a strong claim — make sure your evidence column says WHERE you saw evidence of that in the images.'] After 5 minutes, cold-call 3 students. After each student reads their claim, say to the class: 'Thumbs up if you think the evidence we gathered today supports this claim. Thumbs sideways if you are not sure. Thumbs down if you think today's evidence contradicts this claim.' [COUNT the thumbs visibly — 'I see 12 up, 8 sideways, 2 down — interesting. Why might 2 people say thumbs down?'] After posting 'Our Best Current Explanation', say: 'This paper will change. Every lesson in this unit, we will come back and either confirm, revise, or completely rewrite what it says. Science is not finished — and neither are we.'	CLAIM — EVIDENCE — REASONING (CER): CER is a structured argumentation strategy (NGSS SEP 6: Constructing Explanations; SEP 7: Engaging in Argument from Evidence) that prevents students from restating textbook knowledge and forces them to ground claims in the specific evidence generated in THIS lesson. The public posting of 'Our Best Current Explanation' externalises the class's current model and creates accountability for revision — students can physically see their thinking evolve across the unit. The thumbs voting makes reasoning social and visible without being threatening.	OBSERVABLE INDICATOR: Every student has a written CER in their exercise book. Teacher evaluates: (1) Does the CLAIM name specific structures (e.g., 'cell membrane', 'cytoplasm') or remain vague ('cells have stuff inside')? (2) Does the EVIDENCE column cite the specific images or historical reading rather than general knowledge? (3) Does the REASONING connect to the phenomenon — i.e., does the student gesture toward the fact that red blood cells challenge any claim about nuclei? Students whose CER identifies the contradiction between 'all cells have a nucleus' (common prior belief) and the anucleate red blood cell observation are flagged mentally by the teacher as ready for extension in Lesson 2.
Driving	Each student receives two sticky	Distribute sticky notes and say:	DRIVING QUESTION BOARD	OBSERVABLE INDICATOR:

Question Board (DQB) Creation  VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	notes (or two small pieces of paper if sticky notes unavailable). On sticky note 1 (one colour — e.g., yellow): students write ONE question they still cannot answer about cells after today's lesson. On sticky note 2 (a different colour — e.g., blue): students write ONE prediction they made at the start of class that they now think might be WRONG and why. Students bring their sticky notes to the front and place them on the class Driving Question Board (DQB) — a large sheet of paper or a section of the classroom wall divided into columns: 'Questions about Cell Structure' 'Questions about Why Cells Look Different' 'Questions about How Cells Work'. Teacher helps students categorise their sticky notes into the correct column as they come forward (this takes 3–4 minutes as students arrive in waves). Teacher reads 3–4 questions aloud from the board — choosing the most generative ones — and says: 'These questions are going to drive everything we do for the next 5 lessons. I am not going to answer them now. By Lesson 6, you will answer them yourselves — with evidence.' Teacher adds ONE teacher-generated question to the DQB: 'Why does a red blood cell have NO nucleus — and is that a defect, or is it actually a design feature that makes it better at its job?'	'Two notes. Two minutes. Note one — yellow: your single most burning question about cells that you cannot answer yet. Note two — blue: one prediction you made at the START of class that today's evidence made you doubt. Be honest — changing your mind is what scientists do.' [START 2-MIN TIMER.] As students post notes, circulate and read aloud strong questions as they go up: 'Listen to this question from [student name] — [read question]. Does anyone else have a similar question? Nod if you do.' After all notes are posted: 'This board is alive. It grows every lesson. Every time we answer a question, we cross it off — in green. Every time answering one question gives us a new question, we add it — in red. By the end of this unit, I expect this board to be covered in both green cross-outs and red new questions.' Point to the teacher-added question about the red blood cell nucleus and say: 'This one question — I promise you — is going to unlock everything about how cells become specialised. Hold it in your mind.'	(DQB): The DQB is a metacognitive tool (NGSS SEP 1: Asking Questions) that makes student thinking visible, persistent, and revisable across the unit. By categorising questions into structural, comparative, and functional columns, the teacher scaffolds the epistemological organisation of biology — structure, variation, function — which mirrors the actual architecture of the KICD Cell Biology sub-strand. The two-colour sticky note system (questions vs. revised predictions) ensures that even the Predict phase is formally honoured and tracked, reinforcing that prediction and revision are core scientific practices, not just warm-ups.	Every student has posted two sticky notes. Teacher photographs the DQB at the end of the lesson (or keeps it intact) for comparison in Lesson 6. Diagnostic value: (1) Yellow questions clustering around 'why do cells look different' signal students are already thinking functionally — ready for specialisation content. (2) Yellow questions clustering around 'what is inside a cell' signal students need more structural scaffolding in Lessons 2–3. (3) Blue 'revised prediction' notes that say 'I thought all cells had a nucleus but the red blood cell doesn't' indicate the phenomenon has done its job — the anomaly has registered cognitively.
Model Building Phase  VIDEO: Photosynthesis in organisms Source: Khan	Students return to the cell sketch they drew in Phase 1 (Predict). Using a different coloured pen, they now REVISE their sketch in light of everything observed and discussed in the lesson. This is NOT a copy of a textbook diagram	Say: 'Return to your Phase 1 sketch — the one you drew before I taught you anything. Pick up a [different colour] pen. You are not throwing your first drawing away — you are revising it. Scientists never throw away their first model.'	ITERATIVE MODEL BUILDING (NGSS SEP 2: Developing and Using Models): By returning to and physically revising the Phase 1 sketch using a different colour, students make their conceptual change visible on the page. The	OBSERVABLE INDICATOR: Each student's revised sketch shows visible changes from the Phase 1 sketch (different colour additions, question marks, margin annotations). Teacher evaluates the Model Changelog for evidence

Academy (English - US curriculum) Search ARES for similar videos HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	— students use only what they have built through the lesson's evidence. Revised sketches must include: (a) at least one structure they are NOW confident all cells share, labelled with a sentence explaining WHY they believe it is universal; (b) at least one structure they are now UNSURE about, marked with a '?' and a note explaining the uncertainty; (c) an annotation in the margin: 'This model is INCOMPLETE because I still do not know ____'. Students also write a brief 'Model Changelog' entry (2 sentences) beneath their revised sketch: 'I changed ____ from my original drawing because the evidence showed ____.' Teacher collects books at the end of the lesson (or photographs the page) so that the Lesson 1 model can be compared with the Lesson 6 final model — making growth of understanding visible.	They improve it.' [START 7-MIN TIMER for revision.] Circulate and ask targeted questions to 3–4 students: 'What are you adding that you did not have before — and what evidence from today tells you to add it?' [WAIT 10 SECONDS.] 'What are you leaving as a question mark — and WHY are you not sure?' [WAIT 10 SECONDS.] 'Your sukuma wiki leaf cell has a very thick border. Is that in your model? Should it be in a GENERIC cell model?' [WAIT 15 SECONDS — this seeds the plant vs. animal cell distinction for Lesson 3.] After 7 minutes: 'Write your Model Changelog — two sentences. I changed ____ because ____.' You have 2 minutes.' After 2 minutes, cold-call 2 students to read their changelog aloud. Collect books or photograph the model page. Close the lesson by returning to the projected phenomenon images: 'We started this lesson with these two images. We end this lesson with the same two images — but now you see more than you did 80 minutes ago. Next lesson, we zoom in. We will look at the actual structures inside these cells and ask: what is each one built for?' Do NOT summarise or provide answers — leave the productive tension intact.	'Model Changelog' formalises the process of evidence-based revision — a metacognitive habit that is central to scientific practice and to CBE's emphasis on self-assessment. The deliberate incompleteness of the model ('This model is incomplete because ____') prevents premature closure and sustains curiosity across the unit. Comparing Lesson 1 and Lesson 6 models at the end of the unit provides a powerful, concrete artefact of learning progression for both students and the teacher.	of genuine evidence-based reasoning — 'I changed the nucleus to a question mark because the red blood cell image showed no nucleus' scores higher than 'I changed it because the teacher said so.' EXIT CHECK: Before leaving, each student holds up their book open to their revised sketch — teacher does a rapid visual scan. Any student whose revised sketch is identical to their Phase 1 sketch is individually asked: 'What is one thing from the sukuma wiki image that surprised you today?' to probe whether the observation registered. Lesson 1 model pages are kept intact in books — students are explicitly told: 'Do NOT erase or improve this after class. It is a record of where your thinking is today.'
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. PHENOMENON UPTAKE: Did the blood smear + sukuma wiki images create genuine confusion and curiosity, or did students seem uninterested? If uptake was low, consider: was the silent observation time long enough? Should the images be introduced with a brief narrative hook — e.g., 'Imagine you are the Form 4 student from Kisumu seeing your own blood cells for the first time...'?
2. PREDICTION QUALITY: Review the Phase 1 sketches. What proportion of students drew a nucleus? What proportion drew a cell membrane? What proportion drew organelles they may have encountered informally (mitochondria, chloroplasts)? This data tells you what scaffolding Lessons 2 and 3 must prioritise.

3. DQB RICHNESS: Photograph the DQB and count questions by category. If the majority of questions are in 'Questions about Cell Structure' (structural) rather than 'Questions about Why Cells Look Different' (functional), you may need to build a bridging activity at the start of Lesson 2 that pushes students toward functional thinking earlier.
4. RED BLOOD CELL ANOMALY: Did most groups identify the absence of a nucleus in red blood cells independently, or did it go unnoticed? If fewer than half the groups noticed it, consider adding a zoomed-in inset image of the red blood cell with an arrow pointing to the region where a nucleus would be expected, labelled 'Where is the nucleus?' — for use from Lesson 2 onward.
5. HISTORICAL THREAD: Were students genuinely engaged with the Hooke-to-Virchow reading, or did it feel like an interruption to the phenomenon work? If disconnected, consider reframing the reading as a 'scientist detective file' and asking: 'Which of these scientists would have been most useful to have in the lab with us today?'
6. EQUITY CHECK: Which students did NOT speak during cold-calls or group discussion? Note their names and create deliberate entry points for them in Lesson 2's structured pair work.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed two microscope images side by side: a stained human blood smear showing biconcave anucleate red blood cells and nucleated white blood cells, and a sukuma wiki (kale) leaf cell showing a rectangular shape, chloroplasts, a large central vacuole, and a visible cell wall. Students completed a structured observation table comparing features across three cell types.
What did I learn?	Students learned that cells can differ dramatically in shape, size, and internal structures while still being classified as cells. They were introduced to the historical development of cell theory through Hooke (1665), Schleiden (1838), Schwann (1839), and Virchow (1855), and practised distinguishing between scientific observations and the conclusions drawn from them.
How does this explain the phenomenon?	Students used Claim—Evidence—Reasoning (CER) to argue — based only on today's image evidence — which structures they believe are universal to all cells. They revised their initial 'generic cell' sketch using a different colour to show evidence-based changes, recorded a Model Changelog, and posted questions and revised predictions to the newly created Driving Question Board (DQB).

LESSON 2 (40 minutes): Seeing Cells for the First Time: Microscopy Practical 1

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of this lesson, learners will have directly observed real plant and animal cells under a microscope, recorded accurate biological drawings, and identified visible structural differences between the two cell types, advancing their investigation of the anchoring phenomenon.
Knowledge	Learners will identify visible cell structures including the cell wall, cell membrane, nucleus, cytoplasm, and vacuole in onion epidermal cells and cheek cells as seen under light microscopy.
Skills	Learners will prepare wet mount slides using the correct technique, focus a light microscope under low and high power, produce labelled biological drawings that follow accepted conventions (sharp pencil, scale, labels with ruled lines), and record comparative observations in a structured table.
Attitudes	Learners will demonstrate curiosity and precision as they observe real biological material, appreciate that scientific evidence comes from careful direct observation, and show responsibility in handling laboratory materials safely and hygienically.
Key Inquiry Question	What are the structural differences between plant and animal cells?
Purpose in Storyline	This lesson is the first OBSERVE phase in the evidence-gathering sequence. Learners move beyond predictions made in Lesson 1 and gather direct observational evidence about real cells. The onion cells serve as an accessible plant cell proxy, while cheek cells represent human animal cells — connecting directly to the blood smear phenomenon. New unexplained observations (no green colour in onion, no nucleus visible in some cheek cells, the box-like walls of onion cells) are harvested onto the Driving Question Board to fuel curiosity for Lessons 3 and 4.
Safety Notes	Learners must not share cotton buds between individuals. Each learner uses their own clean cotton bud for cheek cell collection and discards it immediately into a disinfectant beaker (dilute Jik solution) after use. Slides containing cheek cells are treated as biohazardous and placed in a disinfectant tray after observation. Learners wash hands before and after the practical. No learner with an open mouth sore participates in cheek swabbing — they observe a partner slide instead. Glass slides and cover slips are handled with care; broken glass is reported immediately to the teacher and not touched with bare hands.

B. LESSON OVERVIEW

Learners open the lesson by revisiting their Lesson 1 predictions about what a basic cell looks like. They then conduct a hands-on microscopy practical in which they prepare wet mount slides of onion epidermal cells (plant cell) and their own cheek cells (animal cell). Using low and high power magnification, they observe, sketch, and label what they see, then complete a structured comparison table to record differences. The lesson closes with a Driving Question Board update in which learners post new questions arising from structures they observed but cannot yet explain — for example, the rectangular walls of onion cells, the faint or invisible nucleus in cheek cells, and the absence of any green colour in the onion layer used. These unresolved questions serve as the engine for Lessons 3 and 4.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Learners open their notebooks and spend 3 minutes individually reviewing the sketch of a basic cell they drew in Lesson 1. The teacher projects the two anchoring phenomenon images side by side: the stained blood smear (showing red blood cells and white blood cells) and the sukuma wiki leaf cell image. Learners are asked to circle ONE structure on their Lesson 1 sketch that they predict they will be able to see with a light microscope in an onion cell today, and ONE structure they predict they will see in their own cheek cell. Learners share predictions with a shoulder partner using Think-Pair language: 'I predict I will see _____ in the onion cell because _____.'	Teacher projects the anchoring phenomenon images and says: 'Last lesson you predicted what a basic cell looks like. Today we stop predicting and start seeing. But before we look through the microscope, tell me — what do you EXPECT to see in an onion cell versus in your cheek cell? They are both cells, so what might be the same? What might be different? Take 60 seconds in silence to look at your Lesson 1 sketch and circle your prediction.' [WAIT TIME: 60 seconds of silent individual thinking before any pair talk.] After pairs share, teacher cold-calls 3 pairs and records their predictions on the board in two columns: ONION CELL and CHEEK CELL. Teacher says: 'Keep these predictions visible. At the end of the lesson we will return to them and decide — were we right, partially right, or completely surprised?'	Predictions are anchored to the students own prior drawings from Lesson 1, creating cognitive continuity. Recording predictions publicly on the board creates accountability and a reference point for end-of-lesson comparison, reinforcing the PREDICT-OBSERVE-EXPLAIN cycle explicitly.	Teacher scans circled predictions on learner sketches during pair talk. Looks for evidence of recall from Lesson 1 — do learners mention nucleus, cell membrane, cytoplasm? Misconceptions to note: learners who predict they will see chloroplasts in the onion cell (onion epidermal cells lack chloroplasts — this will become a productive surprise). Teacher notes these learners to revisit during observation debrief.
Observe Phase  VIDEO: Experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Learners work in pairs at benches. Each pair has: 2 glass slides, 2 cover slips, iodine solution in a dropper bottle, a fresh onion, forceps or a ruler edge, cotton buds, methylene blue solution in a dropper bottle, tissue paper, and a light microscope. PART A — Onion Epidermal Cell: Learners peel a thin transparent layer from between two onion scales using forceps (the teacher demonstrates this first). The tissue is placed on a slide with one drop of water, flattened, and a cover slip lowered at an angle to avoid air bubbles. One drop of iodine solution is drawn under the cover slip using tissue paper on the opposite side. Learners focus under low power (4x or 10x objective) then switch to	Teacher first demonstrates the onion peel preparation at the front bench, narrating each step: 'I am peeling the inner layer — it must be thin enough to be almost transparent. If it is too thick the light will not pass through and you will see nothing. Watch how I lower the cover slip at an angle to push out air bubbles.' Teacher then circulates during the practical, asking targeted questions at each bench: 'What is the shape of the onion cells — does it remind you of anything you have seen in Kenya? Think of the bricks used to build a nyumba.' At the cheek cell stage, teacher reminds the class firmly: 'Your cotton bud goes in YOUR mouth. No sharing. After use, it goes directly into the Jik	The structured comparison table forces learners to observe with purpose rather than passively looking. The use of iodine (which stains starch and makes cell walls and nuclei more visible in onion cells) and methylene blue (which stains nuclei in cheek cells) makes abstract structures visible, connecting the chemistry of staining to the biology of observation. The brick analogy for the rectangular onion cells connects to a familiar Kenyan construction context, making the plant cell wall concept tangible before it is formally named.	Teacher examines biological drawings during circulation. Key criteria checked informally: (1) Is the drawing in pencil and occupying most of the circle? (2) Are label lines ruled and touching the structure — not arrows? (3) Is the magnification recorded? (4) Are at least two structures labelled per cell type? Teacher identifies one strong drawing from each cell type to display on the document camera during the Explain Phase debrief. Comparison tables are scanned for the key productive observation: most learners will note that the onion cell has a defined rectangular wall and that the cheek cell does NOT — this is the central evidence piece for the lesson.

	high power (40x objective). They observe and draw what they see in a circle representing the field of view, label what they can identify, and write the magnification used. PART B — Cheek Cell: Each learner gently rubs the inside of their own cheek with their own clean cotton bud, smears the material onto their own slide, adds one drop of methylene blue, covers with a cover slip, and observes. They draw and label what they see. Learners then complete a structured comparison table: FEATURE ONION CELL CHEEK CELL — with rows for shape, cell wall visible (yes/no), nucleus visible (yes/no), colour, other observations.	beaker — not on the bench, not on the floor.' Teacher uses a document camera or pre-prepared large diagram to show learners the difference between a focused cell and an air bubble so they can self-correct. If microscopes are limited, teacher ensures rotation so every learner has at least 5 minutes of independent viewing time. Teacher says: 'When you are looking through the eyepiece, I want you to ask yourself three questions: What shape do I see? Can I see a boundary around each cell? Can I see anything inside? You have 12 minutes. Observe first, draw second — do not start drawing until you have spent at least 2 minutes just looking.' [WAIT TIME: Teacher pauses after circulating to the third bench and asks the whole class to freeze and share one surprising observation — 30 seconds of whole-class listening before continuing practical work.]		
Explain Phase  VIDEO: Experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Learners close their microscopes and return to their seats. The teacher displays two reference biological drawings on the board — one of an onion epidermal cell and one of a cheek cell, both correctly labelled with: cell wall, cell membrane, cytoplasm, nucleus, and vacuole (onion only). Learners compare these to their own drawings and self-assess using a 3-point rubric written on the board: (1) I drew the correct shape and size relative to the field of view; (2) I labelled at least 3 structures with ruled lines; (3) I recorded the magnification. Learners circle where they scored well and underline one thing to improve. The teacher then leads a whole-class discussion focused on	Teacher projects the two reference drawings and says: 'Compare your drawing to this. I am not looking for perfection — I am looking for honesty. Where did you get it right? Where do you need to improve? You have 90 seconds.' [WAIT TIME: 90 seconds of silent self-assessment.] Teacher then cold-calls 4 learners to share one difference they recorded in their comparison table. Teacher uses these contributions to build the consensus table on the board, saying: 'Kweli — that is genuine scientific evidence. You observed that the onion cell has a thick boundary that the cheek cell does NOT have. We are going to call that the cell wall. Write that term next to your observation in your	Self-assessment against the reference drawing develops metacognitive awareness of scientific drawing conventions. Returning to Lesson 1 predictions closes the predict-observe loop and makes the cognitive dissonance of surprising results explicit and productive. Naming structures only AFTER observation ensures that vocabulary is anchored to direct experience rather than introduced abstractly — this is a key constructivist move in the storyline.	The self-assessment rubric gives the teacher rapid data on drawing quality without marking every book. Teacher specifically looks for learners who labelled the cell wall on the cheek cell (a misconception — cheek cells have a membrane, not a rigid wall) and addresses this immediately: 'Remember — you did NOT see a thick rectangular wall around your cheek cell. Only the onion cell had that. Correct your label if needed.' This corrects a high-frequency misconception before it solidifies.

	the comparison table: 'What were the biggest differences you recorded between your onion cell and your cheek cell?' Learners share and teacher builds a consensus comparison on the board. Teacher then returns to the Lesson 1 board predictions: 'Which predictions were correct? Which surprised you?' Key surprises are highlighted — especially the absence of green colour in the onion cell (expected by some learners who confused onion with sukuma wiki) and the difficulty of seeing the nucleus clearly in cheek cells. Teacher introduces the formal names for structures observed: cell wall, cell membrane, nucleus, cytoplasm, vacuole. Teacher clarifies: 'Today you observed structures. We have NOT yet explained what each structure DOES. That is the question we carry into Lesson 3 and Lesson 4.'	table.' Teacher asks: 'Did anyone predict they would see green structures in the onion cell?' [Hands raised.] 'And did you see green?' [No.] 'Interesting. So the sukuma wiki cell we saw in Lesson 1 had green structures, but the onion cell does not — even though both are plants. What question does that raise for you?' Teacher pauses and gestures toward the DQB. [WAIT TIME: 20 seconds for learners to formulate a question mentally before the DQB phase begins.]		
Driving Question Board (DQB) Creation  VIDEO: Experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner receives one sticky note (or tears a small piece of paper if sticky notes are unavailable). They write ONE question that arose from today's practical — something they observed but cannot yet explain. Learners are prompted with sentence starters on the board: 'I noticed _____ and I wonder why _____.' or 'I expected to see _____ but instead I saw _____ — why?' Learners post their question notes on the class Driving Question Board under the section labelled LESSON 2: MICROSCOPY EVIDENCE. Teacher reads 5 to 6 questions aloud, clusters related ones together with a marker circle, and identifies which questions will be	Teacher says: 'Scientists do not only record what they see — they record what confused them. That confusion is the engine of science. Write your most burning question from today on your piece of paper. One question only — your most important one. You have 60 seconds.' [WAIT TIME: 60 seconds of independent writing, no discussion.] As learners post their questions, teacher reads them quickly. Teacher then selects questions and says: 'This question — why did the onion cell not have green colour — is connected to our original phenomenon. Remember the sukuma wiki leaf from Lesson 1? We are going to answer this in Lesson 4 when we study plant-specific organelles. I	The DQB technique transforms student confusion into a visible, shared research agenda, making the inquiry process transparent and collaborative. Clustering questions by lesson anticipates the storyline structure, helping learners see that the unit is a coherent investigation rather than a series of disconnected topics. This move also validates the cognitive dissonance learners experienced during the practical as productive scientific thinking.	The DQB questions serve as a window into learner thinking. Teacher notes: (1) Are learners asking questions about structures (observable) or about function (explanatory)? The latter indicates deeper thinking. (2) Are any questions revealing persistent misconceptions from Lesson 1 that were not corrected? These are flagged for follow-up in Lesson 3. Teacher photographs the DQB at the end of class for reference.

	answered in Lessons 3 and 4 versus which will need to wait for later in the unit. Examples of expected learner questions: 'Why did the onion cell not have green colour but the sukuma wiki cell does?', 'What is the jelly-like stuff inside the cheek cell?', 'Why could I not see a nucleus clearly in my cheek cell?', 'Why is the onion cell shaped like a rectangle but the cheek cell is round?', 'Is the thick wall around the onion cell the same as the cell membrane?'	am writing L4 next to this cluster. This question — what is the jelly inside the cheek cell — we will begin to answer in Lesson 3. I am writing L3 next to this cluster.' Teacher says: 'The DQB is now our research agenda. Every lesson from here, we come back to it and answer more questions. By Lesson 8, the goal is to answer the driving question completely — using evidence.'		
Model Building Phase  VIDEO: Experimental probability Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	In the final 5 minutes, learners do a quick individual task: they update their Lesson 1 basic cell sketch using today's observational evidence. They must (a) add or correct at least two structures based on what they actually saw today, (b) write the name of the structure next to each correction, and (c) use a different colour or symbol to mark structures they OBSERVED today versus structures they only predicted in Lesson 1. This updated sketch becomes Version 2 of their evolving cell model, which will be progressively refined through Lessons 3 to 7 until the final 3D model construction in Lesson 7. Learners write the date and the label VERSION 2 — BASED ON MICROSCOPY EVIDENCE at the top of the updated sketch.	Teacher says: 'Your Lesson 1 sketch was your first model — built on prediction alone. Now you have evidence. Scientists revise their models when they get new evidence. That is not weakness — that is science. In the next 4 minutes, update your sketch. Use today as evidence. If you predicted something in Lesson 1 that you did NOT see today, you do not cross it out — you write a question mark next to it, because we may yet find it in a different type of cell.' Teacher circulates and prompts learners who are uncertain: 'What did you actually see in the onion cell that you did not draw in Lesson 1? Start with the shape.' [WAIT TIME: Teacher gives 3 full minutes of independent work before making any whole-class comments.] With 1 minute remaining, teacher says: 'Your model will keep changing. By Lesson 7 you will build a 3D version using materials from around your home. Keep every version — they are evidence of your own thinking growing.'	The iterative model-building strategy makes the nature of science explicit: models are provisional and evidence-driven. Version-labelling creates a longitudinal record of conceptual change that learners can trace across the unit, supporting metacognition. The instruction to mark observed versus predicted structures forces learners to distinguish between empirical evidence and inference — a core science practice.	Teacher collects 5 to 6 updated sketches at the end of class (rotating so that all learners submit once across the unit). Teacher checks: Did the learner add the cell wall to the plant cell and omit it from the animal cell sketch? Is the nucleus now drawn in both cell types but noted as difficult to see in the cheek cell? Are any organelles (such as mitochondria) drawn that were NOT visible today — and if so, does the learner mark them with a question mark, indicating they know the limits of their current evidence? These questions guide the teacher opening of Lesson 3.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Were all learners able to achieve a focused image of at least one cell type? If not, what microscope management adjustment is needed — rotation schedule, demonstration frequency, or peer-support pairing? (2) Which misconceptions appeared most frequently in comparison tables — for example, labelling the cell membrane as the cell wall in cheek cells, or expecting chloroplasts in the onion? Plan a 2-minute correction opener for Lesson 3. (3) What questions on the DQB reveal the richest thinking? Prioritise these in Lesson 3 planning. (4) Did the practical take longer than allocated? If so, consider whether Part B (cheek cells) can be done as a demonstration in future, freeing time for deeper discussion. (5) Were safety protocols followed by all learners, particularly the cotton bud discard procedure? If not, revisit the safety briefing explicitly at the start of Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Onion epidermal cells are rectangular, have a thick outer wall, and show a visible vacuole when stained with iodine. Cheek cells are irregular and rounder in shape, do not have a thick outer wall, and show a faint nucleus when stained with methylene blue. The onion cell did NOT show the green colour seen in the sukuma wiki leaf cell from Lesson 1.
What did I learn?	Both plant and animal cells share a cell membrane and cytoplasm. Plant cells (onion) also have a cell wall that gives them a rigid rectangular shape, and a large vacuole. Animal cells (cheek) lack a cell wall and have a more irregular shape. Structures visible under a light microscope include the cell wall, nucleus, and vacuole — but many internal structures are too small to see at this magnification.
How does this explain the phenomenon?	The rectangular shape of the onion cell is explained by the presence of a rigid cell wall — a structure the cheek cell does not have. The absence of green colour in the onion cell (even though it is a plant cell) raises a new question that will be explained in Lesson 4 when students study chloroplasts and where in a plant they are found.

LESSON 3 (80 minutes): What Is Inside the Cell Doing All the Work? — Organelle Structure & Function (Animal Cell)

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To systematically identify animal cell organelles from electron micrograph images and annotated diagrams, and explain how each organelle's structure enables its specific function.
Knowledge	Learners identify and describe the structure and function of animal cell organelles: nucleus, cell membrane, cytoplasm, mitochondria, ribosomes, rough and smooth endoplasmic reticulum, Golgi apparatus, lysosomes, and centrioles.
Skills	Learners interpret electron micrograph images, annotate biological diagrams, construct factory-analogy concept maps, and use evidence from the anchoring phenomenon to justify organelle-function relationships.
Attitudes	Learners appreciate the remarkable precision and interdependence of cellular structures, and develop curiosity about how microscopic organisation underpins the visible differences observed in the blood smear and sukuma wiki slide from the anchoring phenomenon.
Key Inquiry Question	What are the structural differences between plant and animal cells? How does cell specialisation enable different cells to perform specific functions in living organisms?
Purpose in Storyline	In Lessons 1–2 learners observed the anchoring phenomenon (the blood smear and sukuma wiki leaf slide) and generated DQB questions about WHY cells look different. In Lesson 3, learners now open the 'black box' of the animal cell — they examine what organelles exist inside, what each one does, and begin to build a reference model of a generic animal cell. This prepares them to explain in Lessons 4–5 why specialised cells (e.g., the enucleated red blood cell, the white blood cell with a multi-lobed nucleus) deviate from this generic model.
Safety Notes	No hazardous materials are used in this lesson. If printed electron micrograph images are handled, ensure hands are clean. Remind learners to handle printed materials carefully. If the school has a projector, ensure cables are secured to avoid tripping hazards.

B. LESSON OVERVIEW

Learners begin by revisiting the anchoring phenomenon — the blood smear image showing enucleated red blood cells beside nucleated white blood cells — and reactivate their DQB questions from Lessons 1–2. They predict which organelles must be inside a 'typical' animal cell before any instruction. They then systematically observe a set of annotated electron micrograph images (or high-quality printed diagrams) of animal cell organelles, matching each to its structure and function. The central sensemaking activity uses a Nairobi Industrial Park factory analogy: the cell is a manufacturing factory in the Industrial Area along Mombasa Road, and each organelle maps onto a department in that factory (nucleus = Managing Director's office; mitochondria = generator room; ribosomes = production line workers; rough ER = internal transport corridor; Golgi apparatus = packaging and dispatch department; lysosomes = waste management and security unit; centrioles = site engineers who set up scaffolding during construction). Learners build an annotated concept map linking organelle → structure feature → function → factory-analogy role → connection to the phenomenon. The lesson closes with a return to the DQB where learners post 'answered' sticky notes and add new refined questions, advancing the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown the anchoring	1. Display or distribute the	Elicit-then-Compare Prediction	Observable indicator: Each learner

 VIDEO: Cells and their Organelles - Example 1 Source: CK-12  Search ARES for similar videos  HTML: Animal-like Protists (Flexbook) Source: CK-12  Search ARES for similar readings	phenomenon image on the board (or printed handout): the blood smear showing biconcave red blood cells with NO nucleus alongside white blood cells WITH large, visible nuclei. The teacher poses the predict prompt: 'Before we open up a typical animal cell today — write down in your notebook: What organelles do you think must be inside a working animal cell? Sketch a rough outline and label at least 4 structures you believe are there, and write ONE reason why you think each one must exist.' Learners work individually and silently for 5 minutes, then share with a shoulder partner for 2 minutes. The teacher cold-calls 4 different learners to share one prediction each and records responses on the board under the heading 'Our Predictions — Lesson 3'. Learners are explicitly told: 'We will come back to this board at the end of the lesson and tick off what we got right — and explain what surprised us.'	anchoring phenomenon image. Say verbatim: 'Look again at our blood smear from Kisumu County Hospital. You told me in Lessons 1 and 2 that the red blood cell has NO nucleus. But what about the white blood cell sitting right next to it — it clearly has structures inside. Today we are going to find out exactly WHAT is inside a typical animal cell.' 2. Read the predict prompt aloud and write it on the board. Give learners 5 minutes of silent individual thinking — enforce this silence. 3. After 5 minutes, say: 'Turn to your shoulder partner. Share your sketch and your 4 organelles. You have 2 minutes.' 4. After partner talk, cold-call EXACTLY 4 learners (select different rows/gender): 'Akinyi, tell me ONE organelle you predicted and why.' Record each on the board. WAIT 10–12 seconds after each response before probing. 5. Say: 'Hold onto these predictions — we are scientists today. At the end of the lesson, we will return to this board and evaluate our thinking.'	Routine — By committing predictions to paper before instruction, learners activate prior knowledge and create a cognitive 'anchor point'. Recording predictions publicly on the board creates accountability and a reference point for reflection at the lesson's end, deepening metacognitive awareness of what was learned.	has a sketch with at least 2–4 labelled structures in their notebook before partner talk begins. The teacher scans the room during silent work and notes whether learners are drawing anything at all (engagement check) and whether any learner already uses correct terminology (e.g., 'nucleus', 'mitochondria') — this informs pacing. During cold-calling, the teacher listens for reasoning quality: are learners giving structure-function reasons ('I think there must be something that gives energy because cells do work') rather than purely memorised labels?
Observe Phase  VIDEO: Cells and their Organelles - Example 1 Source: CK-12  Search ARES for similar videos  HTML: Animal-like Protists (Flexbook) Source: CK-12	Learners receive a set of 4 printed A4 resource sheets (or view on screen if projector is available): Sheet 1 — a labelled transmission electron micrograph (TEM) of a generalised animal cell with key organelles visible; Sheet 2 — close-up electron micrograph images of individual organelles (nucleus showing nuclear envelope and nucleolus; mitochondria showing cristae; rough ER showing ribosomes studded on membrane; Golgi	1. Distribute resource sheets. Say verbatim: 'You are now going to do what a real cell biologist does — look at images and describe EXACTLY what you see. Do NOT jump to telling me what you know from memory. Only write what the image shows you.' 2. Model one example with the nucleus: project or hold up the nucleus micrograph. Say: 'I can see a large, dark, roughly spherical structure. It appears to be enclosed by a double	Image-Based Structural Inference (Science & Engineering Practice: Analysing and Interpreting Data) — Learners extract evidence directly from electron micrograph images before receiving any expository information. This mirrors the practice of real cell biologists and ensures learners engage with visual scientific data rather than passively receiving facts. The Evidence Table scaffolds disciplined observation (SEE) distinct from interpretation	Observable indicator: Completed Evidence Table rows for at least 5 of 7 organelles with distinct structural observations (not just 'it is round'). The teacher checks specifically for quality of structural language — are learners describing membranes, shapes, internal features? A learner who writes 'the mitochondria has folded inner membranes that increase surface area' is demonstrating higher-order inference. Flag any pairs who are writing only one-

Search ARES for similar readings	apparatus showing stacked cisternae; lysosome; centriole pair); Sheet 3 — a blank animal cell outline diagram for annotation; Sheet 4 — an Organelle Evidence Table with columns: Organelle Name What I can SEE in the image (structural features) What I THINK it does (from image alone). In pairs, learners examine each electron micrograph image on Sheet 2, record at least 2 observable structural features for each organelle in their Evidence Table (Sheet 4), and make an inference about its function based purely on what they see. They do NOT receive function information yet — they must reason from the images.	membrane. There is a darker dense region inside — I do not yet know what that is. I am writing exactly that.' 3. Circulate the room. At EACH group, ask at least one probing question: 'What does the wavy folded membrane inside the mitochondria SUGGEST to you about its function?' OR 'Why do you think the ribosomes are so TINY compared to the nucleus?' WAIT 10–12 seconds after posing each question before accepting a response. 4. After 15 minutes of pair work, cold-call 5 learners (different from Phase 1) to share one structural observation each for 5 different organelles. Say: 'Otieno, what did you SEE — not what you know — for the mitochondria?' Record on the board. 5. Affirm pattern-noticing: 'Notice how the structure always gives us a clue about the job. We will use that principle all lesson.'	(THINK), building epistemological rigour.	word observations for immediate scaffolding support during circulation.
Explain Phase  VIDEO: Cells and their Organelles - Example 1 Source: CK-12 Search ARES for similar videos  HTML: Animal-like Protists (Flexbook) Source: CK-12 Search ARES for similar readings	The teacher introduces the Nairobi Industrial Park Factory Analogy. Learners are told: 'Imagine a large manufacturing factory in Nairobi's Industrial Area along Mombasa Road — the kind that assembles products, packages them, ships them, and keeps the lights on. This factory IS your animal cell.' The teacher reveals a large analogy mapping poster (or board diagram) one organelle at a time. For each organelle, the teacher provides the structure-function explanation and then asks learners to articulate the factory-department connection themselves before confirming it. The full analogy set is: (1) Nucleus = Managing Director's Office —	1. Introduce the analogy: 'I am going to describe a factory in Nairobi's Industrial Area. Your job is to tell me which organelle matches each department.' Reveal organelles ONE at a time — do NOT show the full list at once. 2. For each organelle: (a) State the structural feature from the electron micrograph. (b) Ask learners to predict the factory match BEFORE you confirm. Say: 'What department does THIS remind you of?' WAIT 12 seconds. Cold-call one learner. (c) Confirm or redirect, then give the full function explanation. 3. After all 10 organelles, say: 'Now annotate your blank diagram	Anchored Analogy Mapping (Science & Engineering Practice: Constructing Explanations) — The Nairobi Industrial Park analogy provides a culturally grounded, concrete schema that learners can map onto abstract organelle functions. Critically, learners are asked to generate the mapping connection themselves before confirmation, making the analogy an active reasoning tool rather than a passive mnemonic. The return to the anchoring phenomenon (enucleated red blood cell) forces learners to apply new knowledge to authentic evidence, consolidating understanding through explanation	Observable indicator 1: Annotated blank diagram (Sheet 3) has all 10 organelles labelled with function and factory role — teacher checks for accuracy during circulation, particularly whether learners correctly distinguish rough ER (ribosomes attached, protein transport) from smooth ER (no ribosomes, lipid synthesis). Observable indicator 2: The 3-sentence written explanation about the enucleated red blood cell. A strong response connects nucleus absence → no DNA → cannot produce proteins or replicate → cell optimised purely for oxygen transport with maximum space for haemoglobin. A weak response

	holds the master blueprint (DNA), sends instructions to all departments; (2) Cell Membrane = Security gate and perimeter wall — controls what enters and leaves the factory; (3) Cytoplasm = Factory floor — the space where all work happens; (4) Mitochondria = Generator room — converts fuel (glucose) into usable electrical energy (ATP); (5) Ribosomes = Production line workers on the factory floor — assemble products (proteins) following the MD's instructions; (6) Rough ER = Internal conveyor corridor with workers attached — transports newly made proteins and modifies them; (7) Smooth ER = Separate chemical processing unit — produces lipids and detoxifies substances; (8) Golgi Apparatus = Packaging and dispatch department — receives, labels, packages, and ships products (proteins/lipids) to correct destinations; (9) Lysosomes = Waste management and security unit — breaks down faulty products, waste, and invading materials using enzymes; (10) Centrioles = Site engineers/scaffold builders — organise the cell's internal skeleton during cell division. After the full analogy is built, learners annotate their blank cell diagram (Sheet 3) with organelle names, one structural feature, one function, and their factory role. Learners then individually answer: 'Using the factory analogy, explain WHY a red blood cell from the Kisumu blood smear has NO nucleus — what department is missing, and what does that mean for the cell's capabilities?'	— Sheet 3. You have 8 minutes. Each organelle needs: name, one structural feature, one function, factory role.' Circulate and check annotations. 4. Then pose the phenomenon-connection question verbatim: 'Using what you now know about organelles — particularly the nucleus — explain WHY the red blood cell in our Kisumu blood smear has NO nucleus. What does losing that organelle mean for the cell? Write 3 sentences in your notebook.' WAIT for all learners to begin writing before circulating. 5. Cold-call 3 learners to share their explanations. Probe: 'If the nucleus is the MD's office, and the red blood cell has no MD's office — what CAN'T that cell do that the white blood cell CAN?' WAIT 10 seconds. Affirm reasoning that connects to DNA, protein synthesis, or cell reproduction.	rather than recall.	merely states 'it has no nucleus' without functional consequences — teacher uses this to identify learners requiring Tier 2 support.
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Driving Question Board (DQB) Creation  VIDEO: Cells and their Organelles - Example 1 Source: CK-12  Search ARES for similar videos  HTML: Animal-like Protists (Flexbook) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board, which has been displayed since Lesson 1. Each learner has two sticky note colours: GREEN = 'I can now answer this question' and ORANGE = 'I have a NEW question from today's lesson.' Learners spend 4 minutes independently reviewing the DQB, selecting at least ONE question they can now partially or fully answer (green sticky note with their written answer summary), and writing at least ONE new question prompted by today's organelle learning (orange sticky note). New questions may include: 'If the sukuma wiki leaf cell also has a nucleus — does it have the SAME type of nucleus as the white blood cell?' or 'Do plant cells have mitochondria too?' or 'Why do some cells have MORE mitochondria than others?' Learners then do a gallery walk of 3 minutes around the DQB to read peers' new orange questions, placing a small dot sticker next to any new question they also want answered. The teacher tallies the top 3 most-dotted new questions and announces: 'These three questions will DRIVE our next two lessons.'	1. Direct learners to the DQB: 'Take 4 minutes — quietly and independently. Find ONE question on our board you can now answer because of today's lesson. Write your answer summary on a GREEN sticky note and place it over or beside that question.' 2. Then: 'Now write at least ONE new question today's lesson sparked for you — on an ORANGE sticky note. This must be a genuine question, not one you already know the answer to.' 3. After posting, announce the 3-minute gallery walk: 'Walk around the board. Read your classmates' new orange questions. Place a dot sticker next to any question YOU also want answered.' 4. After the gallery walk, tally the top 3 most-dotted questions aloud: 'The class most wants to know: [read top 3]. These will anchor Lessons 4 and 5.' 5. Say verbatim: 'Notice how our DQB is growing — not just with answers, but with better, more precise questions. That is what scientists do. The more you know, the better your questions become.'	Driving Question Board Revision (Science & Engineering Practice: Asking Questions and Defining Problems) — The DQB is a living document that makes the progression of learner understanding visible over the unit. Using two sticky note colours distinguishes consolidation (answered questions) from extension (new questions), preventing the misconception that learning means 'having all the answers'. The dot-voting gallery walk democratises the inquiry process and gives the teacher real-time formative data on learner curiosity and conceptual gaps.	Observable indicator: Every learner has posted at least one green and one orange sticky note. The teacher photographs the DQB after the lesson for records. The teacher reads green sticky note answer summaries after class to assess accuracy of self-assessed understanding — particularly whether learners correctly connect organelle function to the phenomenon (e.g., 'The red blood cell has no nucleus = no Managing Director = cannot make new proteins or divide'). Orange questions that show surface thinking ('What is a Golgi?') versus deep thinking ('If the Golgi packages proteins, what happens in a cell where the Golgi is damaged?') are noted for differentiation planning.
Model Building Phase  VIDEO: Cells and their Organelles - Example 1 Source: CK-12  Search ARES for similar videos  HTML:	Learners open their running 'Cell Model Journal' — a dedicated section of their notebook begun in Lesson 1 — and locate their Lesson 1/2 sketched model of 'what we think a cell looks like'. Using a different coloured pen (to show revision), learners now update their model to reflect today's learning. They must: (1) Add and label all 10 animal cell organelles with structural	1. Instruct learners to open their Cell Model Journal to their Lesson 1 sketch. Say: 'Scientists revise their models when they get new evidence. Today you have new evidence — use a different-coloured pen to show that this is a REVISION, not a replacement.' 2. Write the three model tasks on the board for reference. Give learners 7 minutes for model revision. Circulate and check	Cumulative Model Revision (Science & Engineering Practice: Developing and Using Models) — The Cell Model Journal functions as a longitudinal scientific model that accumulates evidence across all 8 lessons. Using a different-coloured pen for each revision makes the history of thinking visible — both to the learner (metacognitive) and to the teacher (formative). Drawing the protein	Observable indicator: Revised model contains all 10 organelles AND the protein synthesis arrow pathway with labels. The comparison text box must include at minimum: (1) Red blood cell has no nucleus — cannot make proteins or divide; (2) Red blood cell has no mitochondria — relies on anaerobic processes; must include haemoglobin as functional adaptation. Teacher uses the 3–4

Animal-like Protists (Flexbook) Source: CK-12 Search ARES for similar readings	annotations; (2) Draw arrows showing the functional pathway of protein synthesis — from nucleus (DNA instruction) → ribosome (assembly) → rough ER (transport) → Golgi (packaging) → cell membrane (export) — labelling each arrow with the factory analogy role; (3) In a text box beside the model, write: 'How is this GENERIC animal cell model different from the RED BLOOD CELL we saw in the Kisumu blood smear? List at least 2 structural differences and explain what each difference means for function.' Learners will continue building this same model in Lessons 4–8 as new evidence accumulates.	specifically: (a) Is the protein synthesis pathway arrow sequence correct? (b) Does the comparison text box include FUNCTIONAL consequences, not just structural observations? 3. After 7 minutes, cold-call 2 learners to share one thing they revised in their model and WHY. Ask: 'What did your Lesson 1 model get wrong or miss? How do you know?' WAIT 10 seconds. 4. Close with the driving question: 'Our model today is of a GENERIC animal cell. But we know from our blood smear that the red blood cell is NOT a generic cell — it has been modified. Next lesson, we will find out what a plant cell looks like and then start comparing ALL of these to our generic model to explain the original puzzle.' 5. Collect or photograph 3–4 Cell Model Journals for formative review before Lesson 4.	synthesis pathway as a functional arrow sequence moves learners from listing organelles to understanding organelles as an integrated, interdependent system.	collected journals to identify: (a) Learners with accurate, detailed models (ready for extension in Lesson 4); (b) Learners whose pathway arrows are in wrong order — reteach sequence in Lesson 4 warm-up; (c) Learners who cannot articulate functional consequences of structural differences — provide Tier 2 support via paired discussion at start of Lesson 4.
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) PREDICTION QUALITY: Were learners' Phase 1 predictions richer than expected, suggesting stronger prior knowledge from Lessons 1–2? Or were most predictions vague (e.g., 'a nucleus and some other stuff'), suggesting the earlier lessons need more scaffolding of organelle vocabulary? (2) ANALOGY ENGAGEMENT: Did the Nairobi Industrial Park factory analogy land — did learners enthusiastically map organelles to departments, or did some learners find the analogy forced or confusing? If learners proposed their own alternative Kenyan analogies (e.g., a matatu sacco, a market stall), note these for use in Lesson 4. (3) PHENOMENON CONNECTION: When asked to explain WHY the red blood cell lacks a nucleus, how many learners went beyond description ('it doesn't have one') to functional consequence ('it cannot synthesise proteins or replicate')? If fewer than 60% achieved the functional explanation, plan a 5-minute reteach warm-up at the start of Lesson 4 using the matatu analogy: 'A matatu with no driver (nucleus) can carry passengers (haemoglobin/oxygen) but cannot navigate new routes or reproduce.' (4) DQB QUALITY: Were the new orange questions showing genuine conceptual depth (e.g., 'If plant cells also have a nucleus — do they have the same organelles as animal cells?') — these signal readiness to advance to Lessons 4–5. Surface questions suggest learners need more processing time. (5) MODEL REVISION: Check collected Cell Model Journals before Lesson 4. Flag learners whose protein synthesis pathways are incomplete or disordered — the sequence nucleus → ribosome → rough ER → Golgi → cell membrane is a threshold concept for all subsequent lessons on secretion and transport in Strands 2 and 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined annotated electron micrograph images and close-up TEM images of individual animal cell organelles (nucleus, mitochondria, ribosomes, rough and smooth ER, Golgi apparatus, lysosomes, centrioles), recording structural features they could
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	directly see in an Organelle Evidence Table before receiving any expository function information.
What did I learn?	Learners identified and explained the structure and function of 10 animal cell organelles, mapping each to a department in a Nairobi Industrial Park factory analogy. Learners traced the protein synthesis functional pathway (nucleus → ribosome → rough ER → Golgi → cell membrane) and connected organelle knowledge back to the anchoring phenomenon by explaining why the Kisumu blood smear red blood cell lacks a nucleus and what functional consequences follow.
How does this explain the phenomenon?	Learners explained that each organelle's structure is directly related to its function — for example, the folded inner membranes (cristae) of the mitochondria maximise surface area for ATP production, just as a generator room is sized to power the whole factory. They explained the anchoring phenomenon observation — the enucleated red blood cell — as a specialisation trade-off: losing the nucleus (Managing Director's office) sacrifices the ability to make proteins or replicate, but maximises internal space for haemoglobin and oxygen transport, a function critical to its role in the bloodstream.

LESSON 4 (80 minutes): Organelle Structure & Function — The Plant Cell: How Is a Sukuma Wiki Cell Like and Unlike a Human Cell?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To explain how plant-specific organelles — the chloroplast, cell wall, and central vacuole — give plant cells their unique structure and enable them to perform functions that animal cells cannot.
Knowledge	By the end of this lesson, the learner should be able to: (a) identify and describe the structure of plant-specific organelles (chloroplast, cell wall, central vacuole, plasmodesmata); (b) explain how each plant-specific organelle relates to a specific function (photosynthesis, structural rigidity, turgor pressure); (c) compare and contrast the structures of a generalised plant cell and a generalised animal cell using a structured comparison table.
Skills	By the end of this lesson, the learner should be able to: (a) construct an accurate, labelled 2D diagram of a plant cell and an animal cell; (b) use evidence from the sukuma wiki leaf observation (Lesson 3) and the blood smear phenomenon (Lesson 1) to justify structural differences between plant and animal cells; (c) apply understanding of turgor pressure to predict what happens to a bean plant cell when water is scarce.
Attitudes	By the end of this lesson, the learner should be able to: (a) appreciate how microscopic structural differences in cells explain everyday observable phenomena such as wilting crops in a Kenyan shamba; (b) demonstrate curiosity and collaboration while constructing and peer-reviewing 2D cell diagrams; (c) value the importance of evidence-based reasoning when comparing cell types.
Key Inquiry Question	1. What are the structural differences between plant and animal cells? 2. How does cell specialisation enable different cells to perform specific functions in living organisms?
Purpose in Storyline	In Lessons 1–3, students gathered evidence about the blood smear phenomenon (RBCs, WBCs) and observed sukuma wiki leaf cells under a microscope, noticing the green structures, thick borders, and a large central sac. In Lesson 4, students now EXPLAIN those observations: they learn the names and functions of plant-specific organelles and use that knowledge to begin building a comparative 2D model of plant vs. animal cells — a direct step toward answering the driving question about why all these structures are called 'cells' yet look so different.
Safety Notes	No chemicals or sharp instruments are used in this lesson. If using wilted vs. turgid bean seedlings as a demonstration, ensure seedlings are handled gently. Wash hands after handling plant material. Ensure diagrams are drawn in pencil first before inking, to allow correction without waste.

B. LESSON OVERVIEW

This 80-minute EXPLAIN-phase lesson helps Grade 10 learners make sense of their Lesson 3 observations of sukuma wiki leaf cells. Students revisit their Lesson 3 observation drawings and use guided direct instruction, a wilting bean seedling demonstration, a structured comparison table, and collaborative 2D model-building to explain the structure and function of the three plant-specific organelles (chloroplast, cell wall, central vacuole). The lesson culminates in students beginning their labelled 2D comparative cell diagrams — a cumulative model they will refine across subsequent lessons. All explanations are anchored to the driving question phenomenon: why do blood cells and sukuma wiki leaf cells look so completely different, yet both qualify as 'cells'?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Plant Cell Structures (Flexbook) Source: CK-12  Search ARES for similar readings	Students open their science notebooks to their Lesson 3 observation drawings of the sukuma wiki leaf cells. Working individually for 3 minutes, they annotate their drawings by circling THREE structures they noticed that they did NOT see in the blood smear from Lesson 1 (the anchoring phenomenon). They write one sentence predicting: 'I think the green structures in the sukuma wiki cell exist because...' and one sentence predicting: 'I think the thick border around the cell exists to...' Students then share their predictions in pairs (Think-Pair-Share, 2 minutes) before the teacher cold-calls 4 pairs to share with the class.	1. Display the anchoring phenomenon split-screen: left side shows the blood smear image (RBCs and WBCs from Kisumu County Hospital); right side shows the sukuma wiki leaf cell image from Lesson 3. Say verbatim: 'Look at both images. You are the same student who donated blood AND who brought sukuma wiki from your shamba. Both slides are on the bench in front of you. What do you see in the leaf cell that is completely absent in your blood cell? Write it. Circle it. You have 3 minutes — go.' [Allow 3 minutes of silent individual work. WAIT TIME — do not speak.] 2. After 3 minutes, say: 'Now turn to your partner. Share your three circled structures and your two prediction sentences. You have 2 minutes.' [WAIT 2 minutes.] 3. Cold-call EXACTLY 4 pairs. For each pair, ask: 'What is ONE structure you circled, and what is your prediction for why it exists?' After each response, say: 'Interesting — hold that prediction. We are going to test it today.' Record student prediction keywords on the board under the heading: 'Our Predictions — We Will Explain These Today.' 4. Do NOT confirm or correct predictions at this stage. The goal is cognitive dissonance and curiosity.	Think-Pair-Share with Annotated Drawing Review. This strategy activates students' own observation evidence from Lesson 3 as the starting point for new explanation, directly linking the OBSERVE phase to the EXPLAIN phase. By asking students to compare their leaf cell drawing to the blood smear phenomenon, the teacher creates productive cognitive conflict that motivates the explanation to follow.	Observable indicator: Each student has at least 3 structures circled in their Lesson 3 drawing and has written 2 prediction sentences in their notebook. Teacher circulates during individual work and scans notebooks. If fewer than 60% of students can identify 3 distinct structures, teacher pauses and points to the projected image, naming one structure (e.g., 'What is this green oval shape?') to scaffold recall before proceeding.
Observe Phase  VIDEO: Photosynthesis in organisms Source: Khan	The teacher displays two bean seedlings side by side: one well-watered (turgid, upright) and one that has been left without water for 48 hours (wilted, drooping). Students observe for 2 minutes	1. Place the two bean seedlings (one turgid, one wilted) on the demonstration bench. Say verbatim: 'These two plants came from the same packet of beans. One is from a well-watered	Concrete Phenomenon Observation + I Notice / I Wonder Table. Using real bean seedlings from a Kenyan farming context (Kericho / Machakos) grounds the abstract concept of turgor pressure	Observable indicator: Students' I Notice / I Wonder tables contain at least one accurate observational statement about the central vacuole size difference between turgid and wilted cells, and at least

Academy (English - US curriculum) Search ARES for similar videos  HTML: Plant Cell Structures (Flexbook) Source: CK-12 Search ARES for similar readings	and record in their notebooks: (a) one physical difference they can see with the naked eye, and (b) a prediction of what is happening INSIDE the cells to cause this difference. Students then watch a 3-minute annotated diagram walkthrough (teacher-drawn on the board OR a pre-prepared ARES platform diagram) showing the internal structure of a turgid vs. a wilted plant cell, focusing on the central vacuole. Students record key observations in a structured 'I Notice / I Wonder' table in their notebooks.	shamba in Kericho. One has not had water for two days — like a farm during a dry spell in Machakos. Observe them carefully. In your notebook, write ONE physical difference you can see, and predict what is happening INSIDE the cells. Two minutes — silent observation.' [WAIT 2 full minutes. Do not narrate. Let students observe.] 2. Cold-call 3 students. Ask each: 'What did you write for what is happening inside the cells?' After each response, say: 'We are going to use today's lesson to find out if you are right.' Write their predictions on the board. 3. Transition to the board diagram. Draw or display a labelled cross-section of a turgid plant cell and a wilted plant cell. Point to the central vacuole in each. Say: 'This large sac — the central vacuole — is full of water in the turgid cell and almost empty in the wilted cell. Remember the large water-filled sac you drew in your Lesson 3 sukuma wiki observation? THAT is this structure.' [Pause 10 seconds — WAIT TIME.] 4. Say: 'In your I Notice / I Wonder table, record: what you notice about the size of the central vacuole in each cell, and write at least one wonder question.' [Allow 3 minutes for recording.] 5. Cold-call 2 students to share their 'I Wonder' questions. Add these to the class Driving Question Board (DQB) parking lot.	in a visible, tangible, culturally relevant observation. The I Notice / I Wonder table is a structured sensemaking tool that separates empirical observation from inference, scaffolding the move from 'what I see' to 'what it means.'	one genuine wonder question. Teacher scans 5–6 notebooks during the 3-minute recording period. Red flag: if students are writing only visual observations about the seedling (e.g., 'one is drooping') and NOT connecting to internal cell structure, teacher says: 'Look at the diagram on the board. What is INSIDE the cell that is different? Add that to your I Notice column.'
Explain Phase  VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum)	Students receive (or access on the ARES platform) a structured 'Plant Cell vs. Animal Cell Comparison Table' with columns: Organelle / Structure Present in Plant Cell? Present in Animal Cell? Function Connection to Our Phenomenon.	1. Say: 'We are now going to NAME and EXPLAIN the three structures you circled in your Lesson 3 drawings. As I explain each one, you will fill in your comparison table WITH your partner. We pause after each	Guided Direct Instruction with Pause-and-Process (Structured Comparison Table). The comparison table is a disciplinary sensemaking tool that makes abstract structural knowledge visible and comparative. By	Observable indicator: Completed comparison table rows are accurate for all three plant-specific organelles. Specifically: (a) Chloroplast — students state 'photosynthesis / makes food using sunlight' as function and connect it

 Search ARES for similar videos  HTML: Plant Cell Structures (Flexbook) Source: CK-12  Search ARES for similar readings	The teacher delivers a focused 15-minute guided explanation of the three plant-specific organelles (chloroplast, cell wall, central vacuole), using the sukuma wiki leaf cell and the wilting bean as anchoring examples throughout. Students complete the comparison table collaboratively in pairs, pausing after each organelle for a 2-minute pair-fill. After the direct instruction, pairs share their completed rows with another pair (pair-to-pair check, 4 minutes). The class then reconstructs a consensus table on the board.	organelle.' 2. CHLOROPLAST (5 min): Draw a chloroplast on the board. Say: 'This is the chloroplast — the green oval you saw in the sukuma wiki leaf cell. It contains a green pigment called chlorophyll. Its function is to capture energy from sunlight and use it to make food for the plant — a process called photosynthesis. This is why sukuma wiki is green. This is why a plant can make its OWN food while your blood cells cannot. Animal cells have NO chloroplasts.' Point to the phenomenon image: 'The student's blood cell — does it make its own food? No. It needs glucose delivered to it. The sukuma wiki leaf cell? It MAKES its own glucose. Same word — cell — completely different strategy.' [WAIT 10 seconds.] Say: 'Fill in the chloroplast row in your table now — 2 minutes with your partner.' [WAIT 2 minutes.] Cold-call 1 pair to read their 'Connection to Our Phenomenon' cell. 3. CELL WALL (5 min): Draw a cell wall on the board outside the cell membrane. Say: 'Every plant cell has a rigid outer layer called the cell wall, made of a tough substance called cellulose. It is like the fence around a shamba — it gives the cell its shape and strength. This is why the sukuma wiki cells you drew looked rectangular and the blood cells looked round and flexible. Animal cells have a cell membrane but NO cell wall.' Ask: 'Why might it be an advantage for a blood cell NOT to have a rigid wall?' [WAIT 15 seconds.] Cold-call 2 students. Expected answer: flexibility to squeeze through narrow capillaries. Say: 'Exactly — the RBC you saw in the blood	pausing after each organelle for a 2-minute pair-fill and connecting every explanation back to the anchoring phenomenon (blood smear vs. sukuma wiki cell), the teacher ensures students are not just receiving information but actively constructing meaning. The pair-to-pair check introduces peer accountability and surfaces misconceptions before whole-class consolidation.	to the sukuma wiki being green; (b) Cell wall — students state 'cellulose / rigidity / shape' and connect it to the rectangular appearance of leaf cells vs. round RBCs; (c) Central vacuole — students state 'stores water / turgor pressure / keeps plant upright' and connect it to the wilting bean. Red flag misconception to listen for: 'The cell wall IS the cell membrane' — address immediately: 'These are two separate structures. The cell membrane is present in ALL cells. The cell wall is an EXTRA layer found ONLY in plant cells, OUTSIDE the membrane.'
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		smear needs to flex and bend. A rigid wall would be a disadvantage. Structure matches function.' [WAIT 10 seconds.] 'Fill in the cell wall row — 2 minutes.' 4. CENTRAL VACUOLE (5 min): Point back to the wilting bean diagram. Say: 'The large sac pushing everything to the side in your sukuma wiki drawing is the central vacuole. In a healthy plant cell, it is full of water and pushes the cytoplasm and organelles outward against the cell wall. This creates TURGOR PRESSURE — the internal pressure that keeps a plant upright and firm. When the central vacuole loses water — as in our wilted bean — turgor pressure drops, the cell becomes flaccid, and the plant droops. Animal cells have small vacuoles, but nothing like this.' Ask: 'If a sukuma wiki plant in your family's shamba in the Rift Valley wilts during a dry spell, what is happening at the cellular level?' [WAIT 15 seconds.] Cold-call 3 students. 'Fill in the central vacuole row — 2 minutes.' 5. Pair-to-Pair Check (4 min): Say: 'Turn to the pair next to you. Compare your tables row by row. If you disagree on any cell, discuss and decide together which answer is more accurate. Be ready to justify.' [WAIT 4 minutes.] 6. Cold-call 3 pairs to contribute one row each to the class consensus table on the board. Correct any misconceptions explicitly.		
Driving Question Board (DQB) Creation  VIDEO: Photosynthesis in organisms	Students individually write one new sticky note (or DQB card on the ARES platform) in response to the prompt: 'What can I NOW explain about the driving question that I could NOT explain at the start of this lesson?' They also	1. Direct students to the DQB (physical board or ARES platform DQB feature). Say: 'You now know three things today that you did not know at the start of the lesson: what a chloroplast does, what a cell wall does, and what a central	Driving Question Board (DQB) — Now I Can Explain / Still Wondering Protocol. This metacognitive sensemaking strategy makes student learning growth visible and keeps the entire lesson sequence tethered to the	Observable indicator: 'NOW I CAN EXPLAIN' cards contain a specific organelle name AND a functional explanation AND a connection to the phenomenon (blood smear or sukuma wiki cell). Vague cards such as 'I learned about plant cells'

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Plant Cell Structures (Flexbook) Source: CK-12 Search ARES for similar readings	write one remaining 'burning question' — something today's lesson raised that they still cannot answer. Students post both cards on the class Driving Question Board under the headings 'NOW I CAN EXPLAIN' and 'STILL WONDERING.' The teacher reads aloud 4–5 cards from each column and uses them to preview upcoming lessons.	vacuole does. Write on your NOW I CAN EXPLAIN card — ONE sentence connecting today's learning to the driving question. For example: NOW I can explain why sukuma wiki cells look rectangular — they have a rigid cell wall made of cellulose. Be specific.' [Allow 3 minutes for writing.] 2. Say: 'Now write your STILL WONDERING card. What question did today's lesson CREATE for you that we have NOT yet answered? It must be a genuine question — not just a repeat of something I already explained.' [Allow 2 minutes.] 3. Students post cards. Teacher reads 4 NOW I CAN EXPLAIN cards aloud, affirming specificity. Then reads 3 STILL WONDERING cards aloud. For each STILL WONDERING card, say either: 'We will answer this in Lesson [X]' or 'This is a great question — let's add it to our board and return to it.' 4. Highlight the overarching driving question on the board. Say: 'We now understand the plant side of the puzzle. Next lesson, we revisit the animal cell — and then we begin to answer the BIG question: what do ALL cells share, despite looking so different?'	driving question. The 'Still Wondering' cards are pedagogically vital: they surface genuine student curiosity and incomplete understanding, which the teacher can use to plan differentiation and sequence upcoming lessons responsively.	are redirected: teacher says, 'Can you name ONE organelle and say what job it does in the sukuma wiki cell?' Red flag: if the majority of 'STILL WONDERING' cards ask basic definitional questions (e.g., 'What is a chloroplast?'), this signals the direct instruction in Phase 3 was not retained and the teacher should open the next lesson with a brief retrieval practice activity before proceeding.
Model Building Phase VIDEO: Photosynthesis in organisms Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML:	Students begin constructing their cumulative 2D labelled diagrams — one diagram of a generalised plant cell and one of a generalised animal cell — in their science notebooks. They use a provided scaffold (a blank outline of each cell type with leader lines for labels) available on the ARES platform. They must label: for the plant cell — cell wall, cell membrane, nucleus, cytoplasm, chloroplast, central vacuole,	1. Distribute or direct students to the ARES platform diagram scaffold. Say: 'You are now going to build your model — a labelled 2D diagram of BOTH cell types. This diagram will grow across the next four lessons as we add more detail. Today, use only what we have learned in Lessons 1–4. Draw in pencil first. Label clearly with leader lines. You have 12 minutes to complete both diagrams.' [Set a visible timer.	Cumulative 2D Model Building with Structured Peer Review. Model-building is a core NGSS Science and Engineering Practice (Developing and Using Models). By making the diagrams explicitly cumulative ('Version 1'), the teacher signals that scientific models are not finished products but living representations of growing understanding. The 4-point peer review checklist operationalises the lesson's	Observable indicator: Completed Version 1 diagrams include all 7 required plant cell labels and all 5 required animal cell labels, with NO cell wall or chloroplast on the animal cell diagram, and the central vacuole drawn large and centrally in the plant cell. Peer review checklists are completed and signed. Teacher collects 6–8 notebooks at the end of class to check diagram accuracy and peer review quality. Any notebook

Plant Cell Structures (Flexbook) Source: CK-12  Search ARES for similar readings	mitochondria; for the animal cell — cell membrane, nucleus, cytoplasm, mitochondria, small vacuoles. Students colour-code plant-specific organelles in green. In pairs, they peer-review each other's diagrams against a 4-point checklist: (1) All required organelles labelled, (2) Plant-specific organelles correctly identified, (3) Labels are accurate (no cell wall on animal cell, no chloroplast on animal cell), (4) Central vacuole is drawn large and central in the plant cell. Students sign their peer's diagram to confirm the review.	Circulate continuously.] 2. During circulation, look for THREE specific errors: (a) students drawing a cell wall on the animal cell — correct immediately: 'Check your comparison table. Does an animal cell have a cell wall?'; (b) students drawing the central vacuole as small — say: 'In a turgid plant cell, the central vacuole occupies up to 90% of the cell volume. Make it BIG — it is pushing everything else to the edges'; (c) students omitting the cell membrane on the plant cell — say: 'The plant cell has BOTH a cell membrane AND a cell wall. The membrane is the thin layer just inside the wall. Add it.' 3. After 12 minutes, say: 'Pencils down. Exchange notebooks with your partner. Use the 4-point checklist on the board to review their diagrams. You have 5 minutes. If you find an error, circle it gently in pencil and write a question — do NOT just cross it out.' [WAIT 5 minutes.] 4. After peer review, say: 'Sign your partner's diagram to confirm you reviewed it. Then take back your own notebook and make any corrections.' [Allow 2 minutes for corrections.] 5. Close the model-building phase by saying: 'This diagram is Version 1 of your cell model. Every lesson from now, you will add to it. By Lesson 8, it will show every major organelle, every specialised cell type, and your full explanation of the driving question. Keep it safe.'	learning outcomes as observable, checkable criteria, embedding formative assessment directly into the student activity. Colour-coding plant-specific organelles in green creates a visual anchor that ties the diagram back to the sukuma wiki phenomenon.	where the animal cell is drawn with a cell wall or the plant cell is missing a cell membrane is flagged for individual follow-up at the start of Lesson 5.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During Phase 3 (Explain), did students make accurate and specific connections between each organelle and the anchoring phenomenon (blood smear + sukuma wiki cell), or did they complete the comparison table as a rote exercise? If the latter, plan a retrieval-practice opener for

Lesson 5 using the phenomenon images. (2) How many 'STILL WONDERING' DQB cards in Phase 4 asked questions that signal misconceptions (e.g., confusing cell wall with cell membrane, or assuming chloroplasts are present in ALL plant cells, not just green parts)? List these misconceptions and address them explicitly at the start of Lesson 5. (3) In Phase 5 (Model Building), what were the three most common diagramming errors during circulation? Were they addressed in real time, or do they persist in the collected notebooks? Plan targeted corrections. (4) Was the wilting bean seedling demonstration (Phase 2) concrete and compelling enough to make turgor pressure intuitive? If students struggled with this concept, consider bringing in a bag of wilted vs. fresh sukuma wiki from a local market at the start of Lesson 5 for a brief tactile recall moment. (5) Did all learners — including those who may have missed Lesson 3 microscope observations — have sufficient prior evidence to engage meaningfully with the Explain phase? If not, what scaffolding can be provided at the start of Lesson 5?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed two bean seedlings — one turgid and upright (well-watered, like a healthy shamba in Kericho) and one wilted and drooping (water-deprived, like a farm during a dry spell in Machakos) — and connected the visible wilting to changes in the central vacuole inside plant cells. Students also revisited their own Lesson 3 microscope observation drawings of sukuma wiki leaf cells, identifying three structures absent in the blood smear from the anchoring phenomenon.
What did I learn?	Students learned that plant cells contain three organelles NOT found in animal cells: (1) Chloroplasts — green, oval organelles containing chlorophyll that capture sunlight to make food via photosynthesis, explaining why sukuma wiki is green and can produce its own glucose while blood cells cannot; (2) Cell Wall — a rigid outer layer made of cellulose outside the cell membrane, explaining why sukuma wiki leaf cells appear rectangular and have a defined shape, unlike the flexible, round red blood cells in the blood smear; (3) Central Vacuole — a large, water-filled sac that creates turgor pressure to keep plant cells firm and upright, explaining why a bean plant wilts when water is scarce and why the large central sac observed in Lesson 3 was pushing organelles to the edges of the sukuma wiki cell.
How does this explain the phenomenon?	Students can now partially explain the driving question phenomenon: the sukuma wiki leaf cell looks rectangular, green, and has a large central sac because it contains a cell wall (cellulose rigidity), chloroplasts (photosynthesis / green pigment), and a central vacuole (turgor pressure / water storage) — all of which are absent in the red blood cells and white blood cells seen in the Kisumu County Hospital blood smear. Despite these dramatic structural differences, both cell types share a cell membrane, cytoplasm, and nucleus (in most cases), confirming that they are both cells — but specialised for completely different jobs. Students have produced Version 1 of their cumulative 2D comparative cell diagrams, which they will continue to develop across Lessons 5–8.

LESSON 5 (80 minutes): Are Bacteria Cells the Same as Our Cells? Prokaryotes vs. Eukaryotes

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To distinguish between prokaryotic and eukaryotic cells by examining structural evidence, and to explain how the absence or presence of membrane-bound organelles relates to cell complexity and function.
Knowledge	Learner knows that: (a) prokaryotic cells lack a membrane-bound nucleus and membrane-bound organelles; (b) eukaryotic cells possess a membrane-bound nucleus and specialised organelles; (c) bacteria are prokaryotes and are smaller in size than eukaryotic cells; (d) prokaryotes may have a cell wall, flagella, pili, and a nucleoid region; (e) all cells — prokaryote or eukaryote — share ribosomes, a cell membrane, cytoplasm, and DNA.
Skills	Learner is able to: (a) compare and contrast diagrams/images of E. coli (prokaryote) and a human cheek cell (eukaryote) using a structured comparison frame; (b) identify at least five structural differences between prokaryotic and eukaryotic cells; (c) revise a cumulative cell model to include a 'prokaryote column'; (d) use evidence to argue why the fermenting uji phenomenon involves prokaryotic rather than eukaryotic organisms.
Attitudes	Learner appreciates that even the simplest cells — lacking a nucleus — are remarkably well-adapted to perform complex biochemical work, and develops curiosity about microbial life in everyday Kenyan foods.
Key Inquiry Question	1. What are the structural differences between plant and animal cells? 2. How does cell specialisation enable different cells to perform specific functions in living organisms?
Purpose in Storyline	Lessons 1–4 established what a 'generic' eukaryotic cell looks like and how specialised animal and plant cells deviate from that model (RBCs, WBCs, sukuma wiki leaf cells). Lesson 5 now confronts students with the deepest structural divide in all of biology: the prokaryote–eukaryote boundary. By revisiting the fermenting uji supporting phenomenon (bacteria causing fermentation), students discover that some cells have NO membrane-bound nucleus at all — shattering any assumption that 'all cells look basically alike.' This prepares students for Lesson 6, where they will synthesise all cell types into a unified explanatory model and address the driving question directly.
Safety Notes	No live specimens or hazardous chemicals are used in this lesson. All observations are image/diagram-based. If a teacher demonstrates uji fermentation using a sealed jar, ensure the jar is not opened near students' faces — fermentation gases may cause discomfort. Remind learners to wash hands if they handle any food specimens. Supervise responsible internet/device use if digital microscopy images are accessed on ARES platform.

B. LESSON OVERVIEW

In this OBSERVE + EXPLAIN lesson, students revisit the fermenting uji supporting phenomenon to surface a new question: what kind of cells are bacteria? Using annotated diagrams and images of E. coli alongside human cheek cells (eukaryote), students systematically compare prokaryotic and eukaryotic cell structures. Through a structured Claim–Evidence–Reasoning (CER) activity, learners identify the defining differences: no membrane-bound nucleus, no membrane-bound organelles, presence of a nucleoid region, cell wall, flagella, pili, and smaller overall size in prokaryotes. They then update their Driving Question Boards and revise their cumulative cell models to include a 'prokaryote column,' explicitly linking back to the anchoring phenomenon — why the cells in a blood smear, sukuma wiki, AND fermenting uji are all called 'cells' yet look so radically different.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Prokaryote Evolution (Flexbook) Source: CK-12  Search ARES for similar readings	Students are shown two stimulus prompts on the ARES platform: (1) a photograph of a bowl of fermenting uji (the supporting phenomenon from earlier lessons) with the caption 'Millions of bacteria are at work in this uji right now,' and (2) the anchoring phenomenon image — the blood smear showing RBCs, WBCs, and the sukuma wiki leaf cell. Students individually sketch in their exercise books what they predict a bacterial cell looks like compared to the human cheek cell they observed in Lesson 3. They annotate their sketch with at least THREE predicted features. Students then share predictions with a shoulder partner (Think-Pair) before a whole-class share-out.	Display the fermenting uji image and say: 'Before we look at any diagrams, I want you to use what you already know. In Lesson 3 you drew the human cheek cell. Now — in your book — sketch what you think a bacterial cell from this uji looks like. You have 3 minutes. Go.' [Allow 3 minutes of silent individual work. Circulate, noting 2–3 interesting or misconception-revealing sketches.] After 3 minutes: 'Now turn to your partner and share your sketch. What did you include? What did you leave out? Discuss for 2 minutes.' [Allow pair discussion.] Cold-call 3 students — choose one who drew a nucleus, one who did not, and one with an unusual feature. Ask each: 'Walk us through your sketch — what made you decide to draw it that way?' [WAIT TIME: 12 seconds after each question before accepting a response.] Record 3–4 key predicted features on the board under the heading 'OUR PREDICTIONS — Bacterial Cell.' Leave these visible throughout the lesson for later comparison.	Think-Pair-Sketch: Students externalise prior knowledge by producing a drawing before instruction. This activates schema, surfaces misconceptions (e.g., 'bacteria have a nucleus like our cells'), and gives the teacher diagnostic information. Predictions remain on the board so students can experience genuine cognitive dissonance when evidence contradicts them.	Teacher circulates during the 3-minute sketch and looks for: (a) whether students include a nucleus in the bacterial cell (a key misconception to address), (b) whether students reference any features from previous lessons (cell membrane, cell wall, organelles). Teacher mentally notes 1–2 students holding the 'bacteria have a nucleus' misconception to cold-call during the Explain phase for targeted correction.
Observe Phase  VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Prokaryote	Students receive (or access on ARES) a two-panel evidence sheet: Panel A shows a labelled diagram of an E. coli bacterium (prokaryote) with the following structures indicated: cell wall, cell membrane, cytoplasm, nucleoid region (circular DNA, NOT enclosed in a membrane), ribosomes (70S), flagellum, pili, and approximate size scale bar (~2 µm). Panel B shows a labelled	Distribute or direct students to the ARES evidence sheet. Say: 'You are now a medical laboratory technologist — like the one in Kisumu who looked at the blood smear in our phenomenon. Your job is to study these two diagrams very carefully and fill in the comparison table. Every feature you can see, write it down. You have 8 minutes working individually — no talking yet.' [Set	Structured Comparison Table (Evidence Sorting): Students sort observable features into categories — shared vs. unique — before the teacher explains the significance. This is an NGSS Science and Engineering Practice (Analysing and Interpreting Data). The physical act of categorising forces students to confront what is truly different rather than passively reading a list. The group	Teacher checks comparison tables during circulation for: (a) correct identification of 'nucleoid region' (not 'nucleus') in E. coli — this is the must-achieve indicator for this phase; (b) whether students have noted the size difference using the scale bars; (c) whether ribosomes appear in BOTH columns. Any student who writes 'nucleus' for E. coli is immediately redirected: 'Look again at the label on that

Evolution (Flexbook) Source: CK-12 Search ARES for similar readings	diagram of a human cheek cell (eukaryote) with: cell membrane, cytoplasm, nucleus (with nuclear envelope), nucleolus, mitochondria, ribosomes (80S), endoplasmic reticulum, and approximate size scale bar (~20 µm). Students complete a structured Comparison Table in their exercise books with the following columns: 'Feature E. coli (Prokaryote) Human Cheek Cell (Eukaryote) Present in BOTH / Only in Prokaryote / Only in Eukaryote.' They must fill in at least 10 features. After completing the table individually (8 minutes), students compare tables with their group of four and reconcile any differences.	a visible timer. Circulate, specifically checking that students are noting the NUCLEOID region vs. NUCLEUS distinction — this is the conceptual heart of the lesson.] At 8 minutes: 'Now, in your group of four, compare your tables. If you disagree on whether a feature is present or absent, go back to the diagram — the diagram is your evidence.' [Allow 4 minutes of group reconciliation.] Bring class back together. Ask targeted questions to 4 different cold-called students: (1) 'Amara — what did you find in the E. coli cell where you would normally expect a nucleus?' [WAIT 12 seconds.] (2) 'Ochieng — how is the nucleoid region DIFFERENT from the nucleus in the cheek cell?' [WAIT 12 seconds.] (3) 'Wanjiku — look at your scale bars. How much bigger is the cheek cell compared to E. coli?' [WAIT 10 seconds.] (4) 'Kamau — both cells have ribosomes. Why do you think that is?' [WAIT 15 seconds.] Write student responses directly onto the class comparison table on the board.	reconciliation step builds scientific argumentation — students must use the diagram (evidence) to resolve disagreements.	structure — does it say nucleus or nucleoid? What do you notice about whether it has a surrounding membrane?' A quick show-of-hands — 'How many found at least 10 features?' — gives class-level data before moving to Explain.
Explain Phase  VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Prokaryote Evolution (Flexbook) Source: CK-12 Search ARES for similar	Students write a Claim–Evidence–Reasoning (CER) paragraph in response to the following prompt displayed on ARES: 'The bacteria that ferment uji (like Lactobacillus) are prokaryotes. A Form 1 student says: "Bacteria cannot be real cells because they have no nucleus — the nucleus is what makes something a cell." Using your comparison table as evidence, write a CER response that either supports or refutes this claim. Your response must: (1) state a clear Claim, (2) cite at least THREE pieces of structural evidence from	Display the CER prompt and say: 'Now it is time to use your evidence to think like a scientist. Read the prompt carefully. You are going to write a CER paragraph — Claim, Evidence, Reasoning. Remember: Evidence comes from the diagram, not from what you feel is true. You have 10 minutes.' [Circulate purposefully. Look for students who are only writing a claim with no evidence — prompt them: 'Where in your comparison table is the proof for that? Point to it.' Also look for students who cite only one piece of evidence — say:	Claim–Evidence–Reasoning (CER) Writing: CER is an NGSS Science and Engineering Practice (Engaging in Argument from Evidence). It requires students to explicitly link observational data to conceptual conclusions, preventing the common error of asserting knowledge without grounding it in evidence. The 'Glow and Grow' peer critique adds an evaluative layer, pushing students to assess the quality of scientific arguments — a higher-order thinking skill aligned to CBC's Communication and	Teacher reads 3–4 CER responses during circulation. Must-achieve indicator: student cites 'no membrane-bound nucleus' and 'nucleoid region' as specific structural evidence — not just 'bacteria are different.' If a student's CER only states the claim with no structural evidence, teacher prompts: 'Your claim is strong — now find three rows in your comparison table where E. coli and the cheek cell are different, and quote them.' Exit check: after the two volunteers share, teacher asks the whole

readings	today's diagrams, (3) provide Reasoning that connects each piece of evidence to your claim, and (4) explain what ALL cells — prokaryote and eukaryote alike — have in common.' Students write individually (10 minutes), then two volunteers share their CER with the class for peer critique using the 'Glow and Grow' protocol (one strength, one suggestion).	'KICD says scientists use multiple lines of evidence. Can you find two more features that support your claim?'] After 10 minutes, ask for two volunteers to read their CER aloud. After each reading, facilitate peer feedback: 'Give one GLOW — something they did well — and one GROW — something they could add or clarify.' [WAIT 10 seconds after asking for glow/grow before accepting responses. Cold-call if no volunteers.] Synthesise on the board: 'So what do ALL cells share, no matter what?' Guide students to articulate: cell membrane, cytoplasm, DNA (in some form), ribosomes. Write these four as the 'Universal Cell Features.' Then say: 'The bacteria in the uji do not need a membrane-bound nucleus to carry out the chemistry of life — their DNA is free in the cytoplasm. This is why we call them PROkaryotes — from the Greek for BEFORE the nucleus.' Write etymology on board. Explicitly connect back to the phenomenon: 'Remember our Kisumu student looking at the blood smear? The RBCs had NO nucleus either — but RBCs are eukaryotic cells that LOST their nucleus as they specialised. That is very different from bacteria, which NEVER had a membrane-bound nucleus to begin with. This distinction matters.'	Collaboration competency.	class by show of hands: 'How many of you included the word nucleoid in your evidence?' This reveals whether the lesson's core concept has landed before moving to model building.
Driving Question Board (DQB) Creation  VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English)	Students return to the class Driving Question Board. Each group receives two sticky notes of different colours: one colour for NEW ANSWERS/EVIDENCE (green) and one colour for NEW QUESTIONS RAISED (orange). Groups discuss for 3 minutes: 'What did today's lesson answer	Distribute sticky notes and say: 'Go to your group. Look at our Driving Question Board. In 3 minutes, decide: (1) Which question on the board can we now answer — or partly answer — using today's evidence? Write the answer on your GREEN note. (2) What NEW question has today's	Driving Question Board (DQB) Update with Colour-Coded Sticky Notes: The DQB is a running public record of the class's growing understanding. Colour-coding answers vs. new questions makes the epistemic progress visible — students can literally see that science generates new questions	Teacher reads all posted sticky notes during the 5-minute discussion. Green notes that correctly cite 'no membrane-bound nucleus' or 'prokaryote vs. eukaryote' as a new answer indicate understanding has been achieved. Green notes that are vague ('bacteria are smaller')

- US curriculum) Search ARES for similar videos HTML: Prokaryote Evolution (Flexbook) Source: CK-12 Search ARES for similar readings	from our DQB? What new questions has it raised?' Groups post their sticky notes on the DQB. The teacher then facilitates a 5-minute class discussion to cluster new questions into themes. Students also update the 'Evidence We Have So Far' section of their individual Science Notebooks with a one-sentence summary: 'Today I learned that...'	lesson raised for you? Write it on your ORANGE note. Go.' [Allow 3 minutes of group discussion. Circulate, prompting groups who are stuck: 'Think about the original phenomenon — the blood smear, the sukuma wiki. Does today's lesson help explain anything new about why those cells look different from each other?'] After 3 minutes, have each group post their notes. Read 2–3 green notes aloud and affirm: 'This group is saying today's lesson answers the question about whether bacteria are the same as our cells — and they are providing the structural reason. Excellent.' Read 1–2 orange notes and say: 'These new questions are exactly what scientists ask. Look — someone is asking whether viruses are prokaryotes or eukaryotes. That is a fantastic extension question. Let us park it here.' Circle 'parked' questions in a designated DQB corner. End by asking: 'Has today's lesson brought us closer to answering our DRIVING question — why do the cells in your body and in sukuma wiki look so different? Thumbs up, sideways, or down.' [WAIT 8 seconds. Scan the room.] Say: 'Hold that thought — we are building toward the full answer in Lesson 6.'	even as it answers old ones. This strategy builds metacognition (students reflect on what they now know vs. what remains unknown) and sustains intellectual momentum across the 8-lesson sequence.	indicate surface-level understanding — teacher makes note to probe these students in Lesson 6. Orange notes that generate meaningful extension questions (e.g., 'Do all bacteria have flagella?' or 'If RBCs have no nucleus, are they prokaryotes?') indicate high engagement and readiness for synthesis. Teacher photographs the updated DQB for reference in Lesson 6 planning.
Model Building Phase VIDEO: Prokaryotic and eukaryotic cells Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Students retrieve their cumulative cell model from Lesson 4 (a three-column comparison chart: Generic Eukaryotic Cell Animal Specialised Cells Plant Specialised Cells). They now add a fourth column: 'Prokaryotic Cell (e.g., Lactobacillus in uji / E. coli).' Using coloured pens or pencils, students annotate this column with: (a) a labelled sketch of a	Say: 'Take out your cumulative cell model from Lesson 4. Today we are going to make it more powerful by adding a prokaryote column. This model is growing lesson by lesson — by Lesson 8, it will help you fully answer our driving question.' [Allow students to retrieve models — 1 minute.] 'First, add a fourth column header: Prokaryotic Cell — and write the	Cumulative Model Revision (NGSS Developing and Using Models): The growing model is the central sensemaking artefact of the entire 8-lesson sequence. Each lesson adds a column or layer, making students' conceptual understanding literally visible and revisable. Adding the prokaryote column forces students to restructure — not just add to —	Teacher checks 5–6 models during the 7-minute sketch period. Must-achieve indicators: (a) nucleoid region is drawn WITHOUT a membrane envelope; (b) cell wall is present in the prokaryote column; (c) no mitochondria or ER appear in the prokaryote column; (d) ribosomes appear in BOTH prokaryote and eukaryote columns. The

 HTML: Prokaryote Evolution (Flexbook) Source: CK-12  Search ARES for similar readings	prokaryotic cell including nucleoid region, cell wall, cell membrane, cytoplasm, ribosomes, flagellum, and pili; (b) colour-coding to show which features are SHARED with eukaryotes (use the same colour as in previous columns) and which are UNIQUE to prokaryotes (a new colour). Students also add a 'Universal Cell Features' row at the TOP of their model, listing the four features all cells share (cell membrane, cytoplasm, DNA, ribosomes). Finally, students write one sentence below their model connecting it to the anchoring phenomenon: 'This helps explain the phenomenon because...'	example: Lactobacillus from fermenting uji. Now sketch and label the E. coli diagram from today's evidence sheet into that column. Use the same colour-coding system we established in Lesson 2 — shared features in BLUE, unique features in RED. You have 7 minutes.' [Circulate. Check that students are drawing the nucleoid region WITHOUT a surrounding membrane envelope — correct any student who draws a membrane around the DNA in the prokaryote column: 'Remember — PRO means before — there is no membrane surrounding the DNA in a prokaryote.'] After 7 minutes: 'Now add the Universal Cell Features row at the very top of your model. What four things do ALL cells — prokaryote and eukaryote — share?' [Cold-call 4 students, one per feature. WAIT 10 seconds each.] 'Finally — the most important part — write one sentence at the bottom: THIS HELPS EXPLAIN THE PHENOMENON BECAUSE... Connect your model to the Kisumu blood smear or the sukuma wiki or the uji. You have 2 minutes.' [After 2 minutes, cold-call 2 students to share their sentence. Write the best class synthesis sentence on the board.]	their existing model, which is the hallmark of genuine conceptual change rather than additive memorisation. The colour-coding system (consistent across all lessons) allows students to instantly see patterns of shared and unique features across all cell types.	phenomenon-connection sentence is a key formative indicator — a sentence like 'This helps explain the phenomenon because the bacteria in the uji are prokaryotes that have DNA but no nucleus, which is completely different from the WBCs in the blood smear that have large visible nuclei' demonstrates full lesson integration. A sentence like 'This explains cells' indicates surface engagement and warrants follow-up in the next lesson.
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) **CONCEPTUAL DEPTH** — Did students consistently and correctly distinguish 'nucleoid region' (no membrane) from 'nucleus' (membrane-bound) in their CER writing and model? If the majority were still conflating these, the Explain phase needs a more explicit direct comparison using a physical analogy (e.g., 'Imagine your DNA is a book. In a eukaryotic cell, that book is locked in a library — the nucleus. In a prokaryotic cell, the book is lying open on a table in the middle of the room — accessible, but not enclosed'). (2) **PHENOMENON CONNECTION** — Did students' DQB contributions and model sentences genuinely link back to the anchoring phenomenon (blood smear / sukuma wiki / uji), or were responses generic? If connections were weak, begin Lesson 6 by re-displaying the anchoring phenomenon image and explicitly asking: 'Where do prokaryotes appear in THIS picture?' (3) **CER QUALITY** — Were students citing specific structural

features as evidence (e.g., 'no membrane-bound nucleus,' 'presence of nucleoid region,' 'no mitochondria') or were they writing vague claims? If CER quality was low, build in a sentence-starter scaffold in Lesson 6: 'My evidence is [specific feature] from [diagram panel], which shows that...' (4) PACE — The lesson has significant content. If Phase 2 (comparison table) ran over time, consider providing a partially pre-filled table in future iterations, leaving only 5 features for students to complete independently rather than 10. (5) EQUITY — Were there students who did not contribute during cold-calls? Review your cold-call list and ensure all students — including those who are quieter or who struggle with English as a medium — are called on with scaffolded questions ('Tell me ONE thing you see that is different between the two cells') rather than open-ended synthesis questions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students examined labelled diagrams of E. coli (prokaryote) and a human cheek cell (eukaryote) on the ARES evidence sheet, and completed a structured 10-feature comparison table identifying features present in both, only in prokaryotes, or only in eukaryotes.
What did I learn?	Prokaryotic cells (e.g., bacteria like Lactobacillus in fermenting uji) have no membrane-bound nucleus and no membrane-bound organelles; their DNA sits free in the cytoplasm in a region called the nucleoid. Eukaryotic cells (e.g., human cheek cells, sukuma wiki leaf cells) have a membrane-bound nucleus and specialised organelles. Despite these differences, ALL cells share four universal features: a cell membrane, cytoplasm, DNA, and ribosomes.
How does this explain the phenomenon?	Using a Claim–Evidence–Reasoning paragraph, students explained that bacteria ARE real cells — despite lacking a nucleus — because they possess all four universal cell features. Students connected this to the anchoring phenomenon by recognising that the bacteria fermenting uji are prokaryotes whose structural simplicity (no nucleus, no membrane-bound organelles) is fundamentally different from the nuclear loss seen in specialised eukaryotic RBCs, and updated their cumulative cell models to include a prokaryote column alongside the generic eukaryotic, specialised animal, and specialised plant cell columns.

LESSON 6 (80 minutes): Cell Specialisation — Why Don't All Cells Look the Same?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To explain how the structure of specialised cells is directly related to the specific function each cell performs in a living organism.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify the structural features of six specialised cells — red blood cells, guard cells, root hair cells, muscle cells, nerve cells, and sperm cells; (b) explain how each structural feature enables the cell to carry out its specific function; (c) relate the abnormal biconcave shape of sickle cells to the disruption of oxygen-transport function.
Skills	By the end of the lesson, the learner should be able to: (a) analyse evidence cards to match structural features to cellular functions; (b) construct a written structure-function argument using scientific reasoning; (c) revise a cumulative cell model to incorporate specialisation as a deviation from the generic cell.
Attitudes	The learner appreciates that diversity in cell structure is not random but purposefully adapted, and develops curiosity about how microscopic structures explain macroscopic health outcomes such as sickle cell anaemia.
Key Inquiry Question	How does cell specialisation enable different cells to perform specific functions in living organisms?
Purpose in Storyline	In Lessons 1–5 students built and refined a generic cell model and gathered evidence on organelle functions. Lesson 6 is the EXPLAIN phase: students now use that foundation to account for WHY specialised cells deviate from the generic model. By the end of this lesson, students can answer the second half of the driving question — 'how does each cell's structure allow it to do its specific job?' — and revisit the anchoring phenomenon (blood smear, sukuma wiki leaf cell, sickle cells) with a full explanatory model.
Safety Notes	No hazardous materials are used in this lesson. If printed evidence cards are laminated, remind students to handle edges carefully. If a brief microscope observation of a prepared slide is included as an extension, follow standard laboratory rules: carry microscopes with two hands, do not touch the objective lens, report any cracked slides immediately to the teacher.

B. LESSON OVERVIEW

Students enter a lesson anchored in the original blood-smear and sukuma wiki phenomenon. Working in groups of four, they receive a set of six Evidence Cards — one per specialised cell (red blood cell, guard cell, root hair cell, muscle cell, nerve cell, sperm cell) — each card containing a labelled diagram, a Kenya-relevant context clue, and a partial function statement. Through a structured 'Structure–Function Match' activity, groups complete the function statements by identifying which structural feature makes the function possible. A whole-class 'Claim–Evidence–Reasoning' (CER) share-out consolidates understanding. The lesson culminates in students returning to the sickle-cell supporting phenomenon to explain — using their new vocabulary — why the crescent shape of a sickle cell disrupts haemoglobin packing and oxygen transport. Students then update their cumulative cell models on the Driving Question Board, adding a 'Specialisation Arrow' showing how generic cells diverge into specialised forms.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are shown two side-by-side images on the board: (A) the	Display the two images and read the prompt aloud. Say: 'You have	Individual Prediction Writing — This strategy activates prior	Circulate and scan notebooks during the 3-minute silent period.

 VIDEO: C4 photosynthesis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	blood smear from the Kisumu county hospital — showing a biconcave red blood cell and a white blood cell with a large nucleus — and (B) a cross-section of a sukuma wiki leaf cell with a visible chloroplast and central vacuole. Each student independently writes a response to the prompt: 'We have studied the generic cell. These cells are NOT generic. List THREE things that are different about each cell and make a prediction: why might each difference exist?' Students record predictions in their science notebooks. No peer discussion yet — this is individual thinking time.	3 minutes of silent thinking time — no talking, no sharing yet. Write down everything you notice that is DIFFERENT from our generic cell model.' START a visible 3-minute timer. Circulate and observe notebooks — do NOT correct or affirm yet. After 3 minutes, cold-call 4 students (aim for gender balance and at least one student who has been quieter in prior lessons): 'Tell me ONE difference you wrote down — just one.' Record each response verbatim on the board in a column labelled 'Our Predictions.' After all 4 responses, say: 'Hold those predictions — we are going to test every single one of them today using evidence cards. By the end of this lesson you will know whether your prediction was right, partially right, or needs revision.' Do NOT confirm or deny any prediction.	knowledge from Lessons 1–5 (generic cell model) and surfaces student misconceptions before instruction. Recording predictions publicly on the board creates accountability and gives students a personal stake in the evidence-gathering that follows.	Look for: (a) students who correctly identify absence of nucleus in RBCs, (b) students who notice the chloroplast or vacuole in the leaf cell. Students who write only vague observations ('they look different') will need targeted prompting in Phase 2. Cold-call responses on the board give a quick whole-class snapshot of prior knowledge depth.
Observe Phase  VIDEO: C4 photosynthesis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Groups of four receive a set of six colour-coded Evidence Cards spread face-down on their desk. Each card contains: (1) a labelled diagram of the specialised cell, (2) a Kenya-relevant context clue (e.g., 'This cell is found in the blood of a marathon runner from Eldoret — it must carry oxygen to muscles for 42 km'), and (3) a partial function statement with a blank: 'This cell can _____ because its structure includes _____.' The six cells are: RED BLOOD CELL (biconcave disc, no nucleus — Kisumu blood donation context); GUARD CELL (bean-shaped pair, chloroplasts — sukuma wiki shamba context); ROOT HAIR CELL (long hair-like extension — bean seedling germinating in Rift Valley soil	Distribute evidence card envelopes before the phase begins. Say: 'You will work as a group of four. Roles today: Card Reader (reads the context clue aloud), Diagram Analyst (describes what they see in the diagram), Recorder (fills in the two-column table for the group), and Devil's Advocate (challenges the group — asks WHY after every suggestion). Rotate roles for each new card.' Allow 12 minutes for group evidence examination. Circulate to every group at least once. Use these specific probing questions when you crouch beside a group: For RBC card — 'If there is no nucleus, what happened to that space? What might fill it instead?' (WAIT 10 seconds before accepting a response.) For	Evidence Card Analysis with Structured Roles — Assigning rotating roles (Reader, Diagram Analyst, Recorder, Devil's Advocate) ensures all four group members engage with every card rather than one student dominating. The two-column STRUCTURE POSSIBLE FUNCTION table scaffolds the structure-function reasoning that is the core scientific practice of this lesson (NGSS Science and Engineering Practice 6: Constructing Explanations). The Kenya-relevant context clues (Eldoret marathon runner, Kisumu blood donation, sukuma wiki shamba, bean seedling, hot sufuria) connect abstract cellular biology to lived experience, making the evidence personally	During circulation, check that the two-column tables contain specific structural vocabulary (e.g., 'no nucleus,' 'elongated,' 'flagellum,' 'many mitochondria') rather than vague terms ('looks weird,' 'has a tail'). Groups that use only vague language need the teacher to model one example: point to the diagram and say, 'What is the NAME of this structure here — do you remember it from our generic cell model?' This links back to Lesson 4 organelle vocabulary. A group is ready for Phase 3 when at least 4 of their 6 rows contain a specific structural feature matched to a plausible function.

	context); MUSCLE CELL (elongated, striations, many mitochondria — Kenyan athlete sprinting context); NERVE CELL (long axon, myelin sheath, dendrites — signal from finger to brain after touching a hot sufuria context); SPERM CELL (compact head, midpiece packed with mitochondria, long flagellum — fertilisation context). Groups flip cards one at a time, read them aloud within the group, examine the diagram, and discuss: 'What structural features do you see? What might they be for?' Each student records observations in their notebook using a two-column table: STRUCTURE POSSIBLE FUNCTION.	guard cell card — 'These cells are in sukuma wiki. What do plants need to let in and out? How might a bean shape help with that?' (WAIT 12 seconds.) For root hair card — 'Imagine you are a water molecule in the soil near a bean seedling in Rift Valley. How does this shape make it easier for the cell to absorb you?' (WAIT 10 seconds.) Cold-call at least 6 different students across all groups during circulations — not just willing volunteers. After 12 minutes, give a 1-minute warning: 'Finish your last card and make sure your Recorder has something written for all six cells.'	meaningful.	
Explain Phase  VIDEO: C4 photosynthesis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Part A — CER Share-Out (12 min): Each group selects ONE evidence card to present to the class using the Claim–Evidence–Reasoning (CER) framework written on the board: CLAIM: 'The [cell name] can [function] because...' EVIDENCE: 'Its structure includes [specific feature], as shown in the diagram by [describe]...' REASONING: 'This feature helps it function because [scientific explanation]...' Each group has 90 seconds to present. Other groups listen and use a 'Plus/Delta' sticky note: Plus = one thing the presenting group got right; Delta = one thing to add or correct. Part B — Sickle Cell Revisit (8 min): Teacher returns to the anchoring phenomenon. Students are shown three images side by side: a normal biconcave RBC, a sickle-shaped RBC, and a diagram of a blocked blood capillary. Students write a 3-sentence CER explanation in their	For Part A: Write the CER scaffold on the board before the phase begins. Say: 'Every group will present ONE card using this exact structure — Claim, Evidence, Reasoning. You have 2 minutes to decide which card and prepare your 90 seconds.' Use a timer visibly. After each group presents, facilitate Plus/Delta: 'Group 2, hold your sticky notes — what is your Plus for Group 1?' Cold-call the Plus from one group and the Delta from a different group. Do NOT editorialize — say: 'Thank you. Group 1, does that Delta change anything about your reasoning?' (WAIT 8 seconds for Group 1 to respond.) Ensure all 6 specialised cells are covered across the 6 groups — if two groups chose the same card, redirect: 'Group 4, Group 2 already took the nerve cell — choose a different one.' For Part B: Project the three sickle-cell images. Say: 'You have studied normal red blood cells. You now	Claim–Evidence–Reasoning (CER) Framework — CER is an explicit argumentation scaffold (NGSS Science and Engineering Practice 7: Engaging in Argument from Evidence) that separates the three components of a scientific explanation. By requiring students to name the specific structural feature as their EVIDENCE and then articulate the mechanistic link as their REASONING, CER prevents the common misconception that correlation equals explanation (e.g., 'it has no nucleus so it carries oxygen' vs. 'the absence of a nucleus creates more internal space, which allows more haemoglobin molecules to pack in, increasing oxygen-carrying capacity'). The sickle-cell revisit is a deliberate transfer task — students must apply the same structure-function logic to an ABNORMAL case, deepening understanding of the norm by examining its disruption.	Collect 6 Plus/Delta sticky notes after Part A — these reveal whether students can identify correct vs. incomplete structure-function reasoning in peers' explanations. For Part B, use a rapid 4-point rubric during cold-call reading: 1 = claim only, no structure named; 2 = claim + structure named; 3 = claim + structure + function linked; 4 = claim + structure + mechanism explained (e.g., 'sickle shape causes cells to stack and block narrow capillaries, reducing oxygen delivery to tissues'). Students scoring 1–2 need a targeted follow-up question: 'You said the shape causes a problem — what physical thing does the sickle shape do inside a blood vessel?' Students scoring 4 can be invited to add their reasoning to the board as the class model answer.

	notebooks answering: 'Why does the sickle shape prevent the red blood cell from doing its job?' Part C — Whole-Class Debrief (5 min): Teacher collects 3 CER responses (cold-called, not volunteered) and reads them aloud. Class votes: 'Does this explanation include a CLAIM? Does it include EVIDENCE from the diagram? Does it include REASONING that links structure to function?' Thumbs up / thumbs sideways / thumbs down for each criterion.	see what a sickle cell looks like. Using ONLY what you know about structure and function, write three sentences. Use the words: shape, haemoglobin, oxygen, capillary.' (WAIT — allow 5 minutes of silent individual writing.) For Part C: Do NOT call on raised hands. Say: 'I am going to read three responses — Amina, Brian, and Wanjiru, please hand me your notebooks.' Read each response aloud neutrally. For each, ask the class to evaluate against the three CER criteria using thumbs. After all three, say: 'What would make the STRONGEST version of this explanation? What is missing?' Cold-call 2 students for additions.		
Driving Question Board (DQB) Creation  VIDEO: C4 photosynthesis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Students return to the class Driving Question Board (DQB) — a large display that has been accumulating sticky notes, arrows, and model revisions since Lesson 1. Each student receives two sticky notes: one ORANGE ('I can now explain...') and one BLUE ('I still wonder...'). Students write one statement on each. They then approach the DQB and place their notes in the designated columns. The teacher facilitates a 3-minute whole-class gallery walk of the new sticky notes. Students look specifically for: (a) a 'still wonder' question that connects cell specialisation to the next topic (Chemicals of Life / Strand 1.4), and (b) any 'I can now explain' note that directly addresses the original driving question. The class votes on the single most important remaining question to carry forward.	Distribute sticky notes and say: 'ORANGE note: complete this sentence — I can now explain [blank] about why cells look different. Be SPECIFIC — name a cell and its feature.' (WAIT 2 minutes of silent writing.) 'BLUE note: What do you still wonder? What does this lesson NOT yet explain about the driving question?' (WAIT 2 minutes.) Direct students to post their notes. During the gallery walk, say: 'Read at least 5 notes that are NOT your own. If you see a BLUE note that matches something you also wonder, put a small tick on it.' After 3 minutes, facilitate a quick vote: 'Which BLUE note asks the most important question we haven't answered yet?' Read the top-voted question aloud and say: 'We will keep this question live on our DQB. In Strand 1.4, when we study the chemicals of life, you may find part of the answer — watch for it.' Write the question on a dedicated 'Carry-Forward' card	Driving Question Board (DQB) with Orange/Blue Sticky Notes — The DQB is a visible, cumulative knowledge artefact. The two-colour system distinguishes consolidated understanding (orange) from productive uncertainty (blue). Publicly posting 'still wonder' questions normalises scientific uncertainty as a driver of inquiry rather than a failure of learning. The tick-voting system reveals shared gaps in understanding that the teacher can use to plan Lessons 7–8 and the transition to Strand 1.4. The Carry-Forward question bridges sub-strands, supporting curriculum coherence.	Read all orange sticky notes before Lesson 7. Categorise them: (A) correctly links a specific structural feature to a specific function — these students are ready for Lesson 7 extension; (B) names a cell but does not name the structural feature — these students need a review prompt at the start of Lesson 7; (C) gives only a generic statement ('I can explain cells') — flag these students for a brief 1-on-1 check-in. Count the proportion of blue 'still wonder' notes that ask mechanistic questions ('HOW does...' / 'WHY does...') versus descriptive questions ('WHAT does... look like'). A class with more than 50% mechanistic blue notes is demonstrating strong scientific thinking and is ready for the model-building phase.

		and pin it prominently.		
Model Building Phase  VIDEO: C4 photosynthesis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Students open their cumulative cell model — begun in Lesson 2 and revised in Lessons 3, 4, and 5. They add a 'Specialisation Layer' to their model. Using a new colour of pen, each student draws a 'divergence arrow' from their generic cell and sketches TWO specialised cells of their choice (from the six studied today), labelling: (a) the structural feature that differs from the generic cell, (b) the function that feature enables, and (c) a Kenya-relevant context sentence (e.g., 'In a sukuma wiki guard cell, the bean shape allows the stomatal pore to open and close, controlling water loss from the shamba plant'). Students who finish early add a third specialised cell and annotate how the sickle cell mutation represents an UNINTENDED structural change that disrupts function. At the end of the phase, students hold up their models for a brief 'gallery share' — they compare their divergence arrows with a partner and discuss: 'Did we choose the same structural feature to highlight? Did we use the same or different Kenya context?'	Say: 'Open your cumulative model. Pick up your green pen — green means SPECIALISATION. You are going to add a Specialisation Layer. You MUST: draw two divergence arrows from your generic cell, sketch two specialised cells, label the structural feature in green, and write one Kenya sentence for each.' Display a model exemplar on the board (teacher-drawn, intentionally incomplete — missing the Kenya sentence) and say: 'My model has a gap — what is missing?' (WAIT 10 seconds — cold-call 1 student.) 'Correct — I have not written my Kenya context sentence. Why does that matter?' (WAIT 8 seconds — cold-call 1 different student.) Allow 10 minutes of independent drawing. At the 8-minute mark, give a 2-minute warning: 'Finish labelling your second cell.' For the gallery share, say: 'Hold your model next to your partner's. You have 2 minutes — find ONE thing that is the same and ONE thing that is different about how you both drew the specialisation. Be ready to share.' Cold-call 2 pairs to share their comparison. Close the phase by saying: 'Your model now shows BOTH what all cells share AND how specialised cells deviate — that is the full answer to our driving question. We will keep refining this model in Lessons 7 and 8.'	Cumulative Model Revision with Divergence Arrows — Adding a 'Specialisation Layer' in a distinct colour makes visible the intellectual progression: the generic cell (built in earlier lessons) is the baseline, and specialisation is a purposeful deviation. The divergence arrow is a visual metaphor for the biological concept of cell differentiation. Requiring a Kenya context sentence for each cell ensures that abstract structural knowledge is always tethered to a real, local organism or body context, supporting long-term retention (NGSS Science and Engineering Practice 2: Developing and Using Models). The partner gallery share introduces low-stakes peer critique, a precursor to the full peer review students will conduct in Lesson 8.	Circulate during the 10-minute drawing period. Check for three specific quality indicators: (1) The divergence arrow originates from the GENERIC cell — not from a blank space (confirms understanding that specialised cells derive from a common template); (2) The structural label uses a specific anatomical term (e.g., 'no nucleus,' 'long axon,' 'flagellum') — not a vague description; (3) The Kenya sentence names a specific organism, location, or context. Students missing indicator 1 need the prompt: 'Show me your generic cell from Lesson 2 — where does specialisation START?' Students missing indicator 3 need: 'Where in Kenya would you find this cell doing its job? Write that place or organism name.' Collect 5 models (randomly selected) at the end of the lesson for written feedback before Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions before preparing Lesson 7:

1. **EVIDENCE CARD ANALYSIS:** Did all groups successfully complete the two-column STRUCTURE | FUNCTION table for all six specialised cells, or were certain cells (e.g., nerve cell, guard cell) consistently incomplete? If yes — redesign the context clue on that evidence card to be more scaffolded, or plan a brief teacher-led example for Lesson 7's warm-up.
2. **CER QUALITY:** When you read the cold-called notebooks during the sickle-cell revisit, what proportion of students reached Level 3 or 4 (structure + mechanism linked)? If fewer than 60% of students reached Level 3, the class needs a 5-minute CER modelling exercise at the start of Lesson 7 before moving to model-building.
3. **DQB BLUE NOTES:** Review the 'still wonder' sticky notes. Are students asking questions that connect cell specialisation to chemistry (e.g., 'What is haemoglobin actually made of?') or to physiology (e.g., 'How does a nerve cell pass a signal — what moves through the axon?')? Questions about haemoglobin chemistry signal readiness for Strand 1.4 (Chemicals of Life). Questions about nerve signal transmission signal readiness for Strand 3.3 (Gaseous Exchange and Respiration / nervous system). Note these for curriculum sequencing.
4. **EQUITY CHECK:** Were the same 4–5 students dominating group discussions during the evidence card activity? If so, restructure groups for Lesson 7, separating dominant speakers. Consider assigning the 'Devil's Advocate' role specifically to quieter students in the next lesson.
5. **KENYA CONTEXT INTEGRATION:** Did students spontaneously use Kenya contexts in their CER explanations and model annotations (e.g., referencing the sukuma wiki shamba, Kisumu hospital, Eldoret marathon runner), or did they revert to generic language? Spontaneous use indicates that the context is becoming genuinely embedded. If students are still using generic language, increase the frequency of Kenya-context prompts in Lessons 7–8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students examined six Evidence Cards, each containing a labelled diagram, a Kenya-relevant context clue, and a partial function statement for one specialised cell (red blood cell, guard cell, root hair cell, muscle cell, nerve cell, sperm cell). They recorded structural observations in a two-column STRUCTURE POSSIBLE FUNCTION table and participated in a CER share-out where each group presented a Claim–Evidence–Reasoning explanation for one specialised cell.
What did I learn?	Students learned that specialised cells deviate from the generic cell model in specific, purposeful ways: red blood cells lack a nucleus to maximise haemoglobin space for oxygen transport; guard cells are bean-shaped to open and close stomatal pores in sukuma wiki leaves; root hair cells are elongated to increase surface area for water absorption in bean seedlings; muscle cells are elongated with many mitochondria to power contraction; nerve cells have long axons to transmit signals rapidly across the body; and sperm cells have a flagellum and mitochondria-rich midpiece to power motility toward an egg. Students also applied structure-function reasoning to the sickle cell supporting phenomenon, explaining that the crescent shape causes cells to stack and block capillaries, reducing oxygen delivery to tissues.
How does this explain the phenomenon?	Students constructed written CER (Claim–Evidence–Reasoning) explanations linking specific structural features of specialised cells to their functions, revised their cumulative cell models by adding a colour-coded Specialisation Layer with divergence arrows, and updated the Driving Question Board with new 'I can now explain' and 'I still wonder' statements. Through these activities, students built a complete explanatory answer to the driving question: cells look different because their structures are precisely shaped to perform their specific jobs, and any disruption to that structure — as in sickle cell anaemia — disrupts function in measurable, life-affecting ways.

LESSON 7 (80 minutes): Building Our Complete Cell Model: Synthesising Evidence to Explain the Anchoring Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students synthesise all prior evidence from Lessons 1–6 to construct annotated 3D cell models using locally available materials, demonstrating understanding of organelle structure-function relationships in both plant and animal cells, and formally revise the Driving Question Board to identify answered and unanswered questions.
Knowledge	Students will know: (a) the names, structures, and functions of all major organelles in plant and animal cells; (b) how specialised cells (red blood cells, white blood cells, sukuma wiki leaf cells) deviate from the 'generic' cell model; (c) which structural features are shared across all cells and which are unique to plant or animal cells.
Skills	Students will be able to: (a) construct an annotated 3D model of a plant or animal cell using locally available materials, justifying each material choice; (b) present and defend their model to peers using precise scientific vocabulary; (c) peer-assess models against agreed criteria; (d) categorise Driving Question Board entries as 'answered', 'partially answered', or 'bridges to future learning'.
Attitudes	Students will appreciate: (a) the value of collaborative model-building as a scientific practice; (b) the creativity and ingenuity involved in representing microscopic structures with everyday Kenyan materials; (c) that science is a cumulative, community process — each lesson's evidence has built toward a richer explanation of the anchoring phenomenon.
Key Inquiry Question	How does each cell's structure allow it to do its specific job, and what does our complete cell model reveal about the similarities and differences between the cells in the anchoring phenomenon?
Purpose in Storyline	This is the culminating model-building and DQB revision lesson of the evidence-gathering sequence. Students have spent Lessons 1–6 gathering evidence about cell structure, organelle function, plant vs. animal cell differences, and cell specialisation. Lesson 7 is the synthesis moment: students now USE all that evidence to build a physical, annotated model that explains the anchoring phenomenon — why the red blood cell, the white blood cell, and the sukuma wiki leaf cell all look so different yet are all called 'cells'. The DQB is formally reviewed, answered questions are celebrated, and remaining questions are explicitly framed as 'bridges' pointing forward to Chemicals of Life (Sub-Strand 1.4), Plant Physiology (Strand 2.0), and Animal Physiology (Strand 3.0).
Safety Notes	No sharp dissection tools are used in this lesson. If scissors are used to cut materials, students must follow standard classroom safety (cut away from body, pass scissors handle-first). Edible materials (avocado seeds, maize grains, mandazi dough, beans) should not be eaten during the lesson as they may have been handled. Wash hands after the activity. Supervise use of glue guns or any adhesives; if hot glue guns are available, only the teacher operates them.

B. LESSON OVERVIEW

In this 80-minute synthesis lesson, student groups of 4–5 are assigned either a PLANT cell or an ANIMAL cell and construct a 3D annotated model using locally sourced Kenyan materials (avocado seed = nucleus, plastic bag = cell membrane, bottle tops = mitochondria, green paper/leaves = chloroplasts, maize grains = ribosomes, dried bean = vacuole, cardboard box frame = cell wall, etc.). Each group prepares a 3-minute presentation explaining: (1) which organelles they included and why, (2) which organelles they deliberately EXCLUDED and why, and (3) how their model explains the specialised cells in the anchoring phenomenon (blood smear + sukuma wiki leaf). After presentations, the class conducts a formal DQB Revision: sorting all DQB cards into 'Answered', 'Partially Answered', and 'Bridge to Future Sub-Strands'. The lesson closes with each student writing a personal 'Science Story' entry connecting their model back to the blood smear phenomenon from the Kisumu hospital.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Cell parts and their functions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Before any building begins, each student individually sketches their PREDICTION of what a complete plant cell OR animal cell model should look like, labelling all organelles they expect to include. They annotate each organelle with: (a) the local material they would use to represent it, and (b) one sentence explaining why that material is a good analogy. Students then do a 'Turn and Talk' with their elbow partner: 'Which organelle do you think will be hardest to represent with a local material, and why?' Students share one remaining question they still have about cell structure that they hope the model-building activity will help them answer.	Teacher displays the anchoring phenomenon image (blood smear from Kisumu hospital + sukuma wiki leaf cell) on the board and says: 'Before we touch any materials today, I want you to THINK. You have spent six lessons gathering evidence. NOW — predict. Sketch your complete cell. Every organelle. Every label. You have 5 minutes. GO.' [Allow 5 minutes of silent individual work. Circulate and note 3 students whose sketches show strong reasoning and 2 students who are missing key organelles — these will be your cold-call targets.] After 5 minutes: 'Turn to your elbow partner. You have 2 minutes. Tell them: which organelle will be hardest to model with local materials, and WHY?' [Allow 2 minutes.] Cold-call 3 pairs using name sticks. Ask: 'Wanjiru and Otieno — what did you decide? And WHY did you choose that analogy?' [WAIT 10 seconds after each response before commenting.] Then ask the whole class: 'Hold up your sketches. Without looking at your neighbour, count the number of organelles you labelled. If you labelled 6 or more, stand up.' [Use this as a quick visual formative check.] Transition: 'Keep your prediction sketch. At the end of today, you will check: was your model accurate? What did you learn that surprised you?'	Individual Prediction Sketch with Analogy Justification — This strategy activates all prior knowledge from Lessons 1–6 and forces students to make their current mental model visible BEFORE instruction. By requiring a local material analogy for each organelle, students must think functionally (e.g., 'the avocado seed is round, hard, and central — like the nucleus which controls the cell and sits in the middle'). This surfaces misconceptions before they are embedded into physical models, allowing the teacher to address them proactively.	OBSERVABLE INDICATOR: Teacher circulates and looks for (a) whether students include the cell membrane AND cell wall in plant cell sketches but only cell membrane in animal cells — a key prior-lesson distinction; (b) whether students can articulate a functional reason for their material analogy, not just a shape match; (c) whether the nucleus is absent from the red blood cell sketch for students who recall Lesson 3's blood smear analysis. Teacher notes names of students missing chloroplasts in plant cell sketches or including a nucleus in the red blood cell — these students will be paired with stronger peers during model-building.
Observe Phase  VIDEO:	Each group of 4–5 receives a 'Materials Kit' bag containing: 1 avocado seed, 1 clear plastic bag,	Distribute material kits and reference cards. Say: 'You are NOT building yet. For the next 8	Material-Organelle Match Table with Justification — This structured observation strategy ensures	OBSERVABLE INDICATOR: Teacher looks at each group's Match Table during circulation.

Cell parts and their functions Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	1 larger ziplock or paper bag, bottle tops (6–8), green paper cut-outs or fresh sukuma wiki leaves (for chloroplasts), dried maize grains (ribosomes), a dried bean or balloon (vacuole), a cardboard sheet (cell wall), string or rubber bands (endoplasmic reticulum), a small sponge or cloth (Golgi apparatus), blue/purple playdough or soil (cytoplasm). Groups also receive a 'Model Reference Card' showing labelled diagrams of a generic plant cell and a generic animal cell (from their textbook or teacher-prepared handout). Groups spend 8 minutes OBSERVING the materials and the reference card together — NOT building yet — and completing a 'Material-Organelle Match Table' where they assign each material to an organelle and write a justification sentence. Groups must also decide: are we building a PLANT cell or an ANIMAL cell, and WHY does that choice matter for our model?	minutes, your job is to be a SCIENTIST planning an experiment. Look at your materials. Look at your reference card. Fill in your Material-Organelle Match Table. Every material must have a home. Every organelle must have a material. And you must agree as a group: plant or animal cell? GO.' [Circulate to all 6 groups within 8 minutes. At each group, ask ONE probing question — choose from: 'Why did you choose the avocado seed for the nucleus and not for the vacuole?', 'Your animal cell — does it have a cell wall? Show me where that decision is in your table.', 'What will you use to show the cell membrane? Is a plastic bag a good analogy? Why or why not?', 'Sukuma wiki cells have chloroplasts. If you are building a leaf cell, how many chloroplasts will you include and where will you put them?'] After 8 minutes, PAUSE the class: 'Hands up — how many groups chose to build a PLANT cell?' [Count.] 'How many chose ANIMAL cell?' [Count.] 'Good. We need both. If all groups chose the same type, I will reassign two groups NOW.' [Reassign if needed.] Say: 'Before you build, each group must do ONE more thing: look at the anchoring phenomenon on the board. Point to the cell in the phenomenon that your model most closely represents. Tell your group WHY.' [Allow 2 minutes of discussion.]	students do not randomly grab materials and start building. By requiring a written justification for every material choice BEFORE construction, students must retrieve and apply functional knowledge of each organelle. The act of matching a physical, tangible Kenyan material to an abstract microscopic structure is a powerful concrete-to-abstract bridge (Bruner's enactive-iconic-symbolic progression). It also surfaces group disagreements ('Should the plastic bag be the cell membrane or the cytoplasm?') that generate productive scientific argumentation.	Green flag: groups correctly distinguish that chloroplasts and cell wall are EXCLUDED from animal cell models and that the large central vacuole is plant-specific. Red flag: groups assign the avocado seed (nucleus) to 'vacuole' in plant cells, or include a cell wall in their animal cell — teacher intervenes immediately with the question: 'What is the FUNCTION of the cell wall? Does an animal cell need that function? Why or why not?' Groups that cannot answer are directed back to their Lesson 2 notes before building begins.
Explain Phase  VIDEO: Cell parts and their functions Source: Khan	Groups spend 20 minutes constructing their 3D annotated models. Each organelle must have a handwritten label flag (on a small piece of paper attached with a	Announce building time: 'You have 20 minutes to build. Every organelle needs a label flag. Every label flag needs three things: name, function, material choice	Annotated 3D Model Construction + Structured Presentation Protocol — The physical act of building a 3D model with local Kenyan materials (enactive learning)	OBSERVABLE INDICATORS during building: (1) Every label flag is present and contains all three required elements — teacher checks this during circulation; any

Academy (English - US curriculum) Search ARES for similar videos HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	toothpick or string) stating: (1) organelle name, (2) function in one sentence, and (3) the material used and WHY it was chosen. After construction, each group delivers a 3-minute structured presentation to the class following a required format: (a) 'Our cell is a [plant/animal] cell because...'; (b) 'The most important organelle in our model is [X] because in the anchoring phenomenon, it explains [Y]...'; (c) 'We deliberately did NOT include [organelle] because...'; (d) 'If our cell were specialised like the [red blood cell / white blood cell / sukuma wiki leaf cell] in the phenomenon, we would change our model by...'. After each presentation, two peer groups provide feedback using a Peer Assessment Rubric (accuracy of organelle placement, quality of justification, connection to anchoring phenomenon, clarity of labels). The presenting group responds to at least ONE piece of feedback before the next group presents.	justification. I will be circulating. I will ask one person at random in your group to explain ANY label on your model — so EVERYONE must understand EVERY organelle. Begin.' [Set a visible timer for 20 minutes. Circulate every 3–4 minutes.] During building, use cold-call probing: approach a group and point to any label flag, then address a student who has NOT spoken yet: 'Akinyi — explain to me why your group chose bottle tops for the mitochondria.' [WAIT 12 seconds.] If the student cannot answer, say: 'Take 30 seconds with your group — then I will come back to you, Akinyi.' [Return in 90 seconds.] At the 15-minute mark, give a '5-minute warning': 'Five minutes. Every label flag must be on your model. No exceptions.' For presentations: use a visible timer showing 3 minutes per group. After each presentation, say: 'Peer assessors — Group [X] and Group [Y] — you have 90 seconds. Written feedback on your rubric. GO.' [After written feedback:] 'Group [Z], read out the most useful piece of feedback you received and tell us how you would change your model based on it.' [WAIT 10 seconds.] Ensure all 6 groups present. If time is tight, combine peer feedback into a whole-class gallery walk (see Phase 4 transition).	consolidates declarative knowledge into embodied understanding. The required presentation format is a sensemaking scaffold: by forcing students to explicitly state what they EXCLUDED and WHY, and to connect their model to the anchoring phenomenon, the protocol prevents surface-level model-making (just assembling objects) and demands deep structural-functional reasoning. The peer assessment feedback loop creates a community of scientific discourse where models are treated as scientific arguments that can be critiqued and improved — mirroring the practice of real scientists.	missing element is flagged immediately. (2) Animal cell models do NOT contain cardboard cell wall or green paper chloroplasts — teacher corrects this before presentation. (3) During the structured presentation, students correctly state at least ONE connection to the anchoring phenomenon (e.g., 'The red blood cell in the blood smear from Kisumu has no nucleus, so if we were modelling that specialised cell, we would remove the avocado seed from our model'). OBSERVABLE INDICATOR during peer assessment: Peer feedback is specific and references the rubric criteria — vague praise ('Good model') is redirected by the teacher: 'That is an opinion, not an assessment. Look at your rubric. Which criterion are you commenting on, and what is your EVIDENCE?'
Driving Question Board (DQB) Creation VIDEO: Cell parts and their functions Source: Khan	The full DQB from Lesson 1 is displayed. All DQB cards from Lessons 1–6 are visible. The class works together in a whole-class discussion to sort every DQB card into one of three categories, physically moving cards into labelled columns: (1)	Display the DQB prominently. Say: 'This board has been growing since Lesson 1. Every question you asked about the student from Kisumu, about the blood smear, about the sukuma wiki leaf — it is all here. Today, we are scientists doing a REVIEW. We are asking:	DQB Tripartite Sort with Dot Voting and Structured Debate — The Driving Question Board revision is a metacognitive sensemaking strategy: students are not just reviewing content but reviewing the STRUCTURE of their own growing knowledge. The dot-voting	OBSERVABLE INDICATOR: Teacher checks that the following key questions are correctly categorised: 'Why does the red blood cell have no nucleus?' should be in ANSWERED (Lesson 3 evidence on cell specialisation — loss of nucleus

Academy (English - US curriculum) Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	ANSWERED — we have clear evidence from our lessons; (2)  PARTIALLY ANSWERED — we have some evidence but there are gaps; (3)  BRIDGE TO FUTURE LEARNING — this question points to a future sub-strand (students must identify WHICH sub-strand: Chemicals of Life 1.4, Plant Physiology 2.0, or Animal Physiology 3.0). Each student receives three coloured sticky dots (green = answered, orange = partial, blue = bridge) and votes by placing dots on cards. Disputed cards are debated briefly (2–3 minutes maximum per dispute). Students then individually write in their science notebooks: 'The question I am most excited to answer in a future lesson is [X] because it connects to the phenomenon by [Y].'	what do we NOW know? Take your sticky dots. Walk to the board. Place your green dot on any card you believe we have FULLY answered with evidence. Orange if partial. Blue if it is a bridge to a future sub-strand.' [Allow 5 minutes for voting. Circulate and note any surprising votes — e.g., if students mark 'How does the cell membrane control what enters the cell?' as fully answered when it is only partially addressed.] After voting, draw attention to 2–3 hotly contested cards: 'This card has both green and blue dots. That means some of us think this is answered and some think it is a bridge. Let us debate. Otieno — you put a green dot. Make your case.' [WAIT 10 seconds.] 'Fatuma — you put a blue dot. Respond.' [WAIT 10 seconds.] After the debate, take a class vote: 'Hands up for green. Hands up for blue.' Move the card to the majority column. Point to all BRIDGE cards: 'These are not failures. These are GIFTS to your future self. Each bridge card is a question that will be answered in Chemicals of Life, Plant Physiology, or Animal Physiology. Write down which sub-strand you think will answer each bridge card in your notebook.' [Allow 3 minutes.] Cold-call 3 students to share one bridge card and its future sub-strand destination. WAIT 12 seconds after each.	system makes individual thinking visible and creates natural disagreement that drives productive scientific argumentation. By explicitly labelling unanswered questions as 'bridges' rather than failures, this strategy builds scientific identity — students see that not knowing is not a deficit but a direction. Connecting bridge questions to specific future sub-strands shows students that the curriculum is a coherent, connected inquiry, not isolated topics.	increases oxygen-carrying capacity). 'What chemicals make up the cell membrane?' should be in  BRIDGE (→ Sub-Strand 1.4 Chemicals of Life). 'How does water move in and out of the sukuma wiki leaf cell?' should be in  BRIDGE (→ Strand 2.2 Transport). Teacher flags any card incorrectly categorised and uses it as a teaching moment: 'You placed this in ANSWERED. What is your evidence from our lessons? Can you cite which lesson gave us that evidence?' If a student cannot cite lesson evidence, the card is moved to PARTIAL.
Model Building Phase  VIDEO: Cell parts and their functions	Each student writes a 'Science Story' entry in their notebook (minimum 8 sentences) that synthesises the entire unit. The Science Story must: (1) Retell the anchoring phenomenon in their	Say: 'You have 12 minutes. In your science notebook, write your Science Story. This is not a summary — it is a STORY. It has a beginning (the student from Kisumu and the mystery), a middle	Science Story Writing with Personal Diagram Revision — The Science Story is a narrative sensemaking strategy that leverages the power of storytelling (a culturally resonant mode in	OBSERVABLE INDICATOR for Science Story: Teacher checks that all five required elements are present. Minimum passing threshold: student correctly states at least TWO structural features

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Cell Transport (Flexbook) Source: CK-12 Search ARES for similar readings	own words (the student from Kisumu, the blood smear, the sukuma wiki); (2) Explain what the red blood cell, white blood cell, and sukuma wiki leaf cell have IN COMMON (shared cell features); (3) Explain what makes each cell DIFFERENT and how those differences link to function; (4) Reference their group's 3D model — what was the best analogy they used and why?; (5) Identify one thing they found surprising and one question they still have. Students then do a final revision of their personal cell diagram (begun in Lesson 1) — annotating it with all new knowledge and explicitly marking which features are 'shared by ALL cells' vs. 'plant cell only' vs. 'animal cell only' vs. 'specialised cell adaptations'. The lesson closes with a whole-class 'Celebration of Learning': teacher reads out 3 anonymised Science Story excerpts that demonstrate exceptional reasoning, and students applaud each one.	(what you discovered in Lessons 1–7), and an end (what you now understand and what questions remain). Minimum 8 sentences. Go.' [Set a visible timer for 12 minutes. Circulate silently — do not interrupt writing. Note 3 strong excerpts to read aloud anonymously at the close.] At 10 minutes: '2 minutes remaining — finish your story and move to your personal cell diagram. Update it NOW with everything you know.' At 12 minutes: 'Pens down. I am going to read three excerpts from Science Stories written today. I will not say whose story it is. When I am done reading each one, I want you to identify: what is the strongest piece of scientific reasoning in that story? Ready?' [Read 3 excerpts. After each, cold-call 1 student: 'Kamau — what was the strongest reasoning in that story?' WAIT 10 seconds.] Close the lesson: 'The student from Kisumu who donated blood at the county hospital gave us a mystery in Lesson 1. Today, YOU solved it. You built the model. You explained the differences. And the questions that remain — those are your bridges to the next great inquiry. Well done.' [Direct students to add their bridge questions to the inside cover of their notebooks as a 'Future Inquiry' list.]	Kenyan oral traditions) to consolidate scientific understanding. Writing a coherent narrative forces students to sequence evidence, identify causality (structure → function), and distinguish what they know from what they are still uncertain about. The personal diagram revision is a visual-spatial consolidation activity that makes conceptual growth visible — students can literally SEE how their diagram from Lesson 1 has grown. The 'Celebration of Learning' close is an affective sensemaking ritual that reinforces scientific identity and community.	shared by all cells (cell membrane, cytoplasm, DNA/genetic material) AND correctly explains at least ONE structural difference between their assigned cell type and a specialised cell from the phenomenon. OBSERVABLE INDICATOR for diagram revision: The revised diagram must show clear visual distinction between 'shared features' and 'cell-type-specific features' — teacher circulates and stamps or initials diagrams that meet this standard before the lesson ends. Students whose diagrams do not meet the standard are given a specific revision task: 'Add a colour key to your diagram: blue = shared by all cells, green = plant cell only, red = specialised adaptation. Complete this before next lesson.'
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. MODEL ACCURACY: Did student groups correctly exclude cell walls and chloroplasts from animal cell models, and correctly exclude centrioles from plant cell models? If groups made systematic errors, what gap in Lessons 1–6 does this reveal?

2. PHENOMENON CONNECTION: During structured presentations, did students spontaneously connect their models to the anchoring phenomenon (the Kisumu blood smear and sukuma wiki leaf cell), or did they need prompting? If prompting was required, consider whether future lessons should make the phenomenon connection more explicit at each phase.
3. DQB QUALITY: Were students able to articulate WHY a bridge question belongs to a specific future sub-strand (e.g., 'How does the cell membrane control what enters?' → Chemicals of Life)? If not, the connections between sub-strands may need to be made more explicit in your lesson planning.
4. EQUITY AUDIT: Which students did NOT speak during presentations or DQB discussion? Were there gender patterns (e.g., girls deferring to boys during group building)? Plan specific cold-call targets for Lesson 8.
5. SCIENCE STORY QUALITY: Did students' Science Stories demonstrate genuine narrative coherence — a beginning (mystery), middle (evidence), and end (explanation) — or did they list facts without causal reasoning? If the latter, consider modelling a Science Story paragraph at the start of Lesson 8's review.
6. MATERIAL ANALOGIES: Which locally available material generated the richest discussion about why it was (or was not) a good analogy? Note this for future years — strong analogies are pedagogical gold worth preserving in your resource bank.
7. BRIDGE QUESTIONS: Review the bridge questions your class identified. Do they align with the actual content of Sub-Strands 1.4, 2.0, and 3.0? If students identified bridges you had not anticipated, consider how to honour those questions explicitly in future lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed their group's material kit (avocado seeds, bottle tops, plastic bags, maize grains, green paper, dried beans, cardboard, etc.) alongside a reference card showing labelled plant and animal cell diagrams. They observed other groups' completed 3D models during structured presentations and peer assessment. They observed the full Driving Question Board from Lessons 1–7 during the DQB Tripartite Sort.
What did I learn?	Students learned how to synthesise structural and functional knowledge of all major organelles into a single coherent 3D model. They learned that plant cells uniquely contain cell walls, chloroplasts, and a large central vacuole, while animal cells lack these but may contain centrioles. They learned that specialised cells (red blood cells, white blood cells, sukuma wiki leaf cells) are DEVIATIONS from a generic cell model, with specific structures added or removed to optimise a particular function. They learned to distinguish 'answered' from 'partially answered' from 'bridge' questions — understanding that unanswered questions are the starting point for future scientific inquiry.
How does this explain the phenomenon?	Students explained why the red blood cell in the Kisumu hospital blood smear has no nucleus (loss of nucleus maximises haemoglobin space for oxygen transport — a specialisation away from the generic cell model). They explained why sukuma wiki leaf cells are rectangular, have thick green chloroplasts, and a large central vacuole (plant cell features: cell wall for rigidity and support, chloroplasts for photosynthesis, vacuole for turgor pressure and water storage). They explained what ALL cells in the phenomenon share: a cell membrane, cytoplasm, and genetic material — the irreducible minimum of a 'cell'. They explained how their 3D model, built from Kenyan everyday materials, represents a 'generic' cell that is the reference point from which all specialised cells deviate.

LESSON 8 (40 minutes): Driving Question Synthesis: So Why DO Cells Look So Different?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students will synthesise all learning from Sub-Strand 1.3 to construct a complete, evidence-based written explanation answering the driving question, using the anchoring phenomenon as the organising frame.
Knowledge	Students will demonstrate that: (a) all cells share core structures such as a cell membrane, cytoplasm, DNA, and ribosomes; (b) plant cells uniquely possess a cell wall, chloroplasts, and a large central vacuole while animal cells may possess centrioles; (c) specialised cells are deliberate deviations from the generic cell model, with organelles added, removed, or modified to optimise a specific function; and (d) prokaryotic cells differ from eukaryotic cells in lacking membrane-bound organelles and a true nucleus.
Skills	Students will practise scientific writing by constructing a structured, evidence-referenced explanation; they will use a consensus class model to verify and refine individual reasoning; they will evaluate the quality of their own and peers explanations against a co-constructed success criteria.
Attitudes	Students will appreciate that scientific understanding accumulates through sustained inquiry rather than single observations; they will show intellectual honesty by acknowledging what remains unexplained and framing those gaps as bridges to future learning.
Key Inquiry Question	Why do the cells in my body and in the sukuma wiki on my plate look so different from each other, and how does each cell structure allow it to do its specific job?
Purpose in Storyline	This is the final consolidation lesson. All eight lessons have built toward this moment: students return to the original anchoring phenomenon — the blood smear and the sukuma wiki leaf cell — and use every piece of evidence gathered across the unit to write a coherent, complete answer to the driving question. The DQB is formally closed, unanswered questions are named as bridges to Sub-Strand 1.4, and the teacher previews the next inquiry arc.
Safety Notes	No practical work is conducted in this lesson. If students handle printed images or cards, no special safety precautions are required. If the lesson takes place in a laboratory, students should remain seated and should not handle any chemicals or equipment.

B. LESSON OVERVIEW

Lesson 8 is a 40-minute synthesis and assessment lesson. In the Predict phase, students revisit the original anchoring phenomenon images and re-examine their Lesson 1 predictions, noting what they now know versus what they originally guessed. In the Observe phase, students examine a final annotated comparison display of the three key cells from the phenomenon — red blood cell, white blood cell, and sukuma wiki leaf cell — and identify the specific structural features that make each cell specialised. In the Explain phase, each student independently writes a full evidence-based explanation answering the driving question in four parts: shared structures, plant-versus-animal differences, specialisation logic, and one Kenya-relevant example. The class then co-constructs a final consensus model on the board. In the DQB Closure phase, all remaining questions are reviewed and labelled as answered, partially answered, or bridged to future sub-strands. The lesson closes with a teacher preview of Sub-Strand 1.4 Chemicals of Life.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Cell parts and their functions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	Students enter to find their original Lesson 1 prediction sketches returned to their desks alongside the anchoring phenomenon displayed on the board: the blood smear image showing biconcave red blood cells and lobed white blood cells next to the sukuma wiki leaf cell image showing rectangular cells with chloroplasts and a large vacuole. Students spend 4 minutes silently comparing their original sketch to what they now know. They annotate their original sketch in a different colour, writing brief notes beside their original predictions to mark what was correct, what was wrong, and what they had not yet known to draw at all. Students are then asked to turn to a partner and complete one sentence each: 'The biggest thing I did not understand in Lesson 1 that I understand now is...' Teacher takes three or four brief volunteer responses to surface the arc of learning before moving to the observation phase.	Display the two anchoring phenomenon images prominently side by side. Return original Lesson 1 sketches to students beforehand. Open with the prompt: 'Look at what you drew on Day 1. You did not know what you did not know. That is exactly where science starts. Take 4 minutes in silence to annotate your original sketch with what you now know.' Allow exactly 4 minutes of silent individual work. Do not circulate and prompt during this silence — allow genuine personal reflection. After 4 minutes, say: 'Turn to your partner. Finish this sentence: The biggest thing I did not understand in Lesson 1 that I understand now is...' Allow 90 seconds of partner talk. Then cold-call three or four students for whole-class sharing. Use affirming but probing responses: 'You said you now understand why the red blood cell has no nucleus — can you say in one sentence what that allows the cell to do?' WAIT TIME: after cold-calling, wait a full 5 seconds before prompting. Do not paraphrase the student answer — repeat it verbatim and ask: 'Does everyone agree with that statement?'	Returning original predictions activates metacognitive reflection. Students see their own conceptual growth made visible, which motivates genuine synthesis rather than rote recall. Partner talk before whole-class sharing lowers the affective barrier for students who are uncertain.	Teacher listens to partner conversations and whole-class responses to identify any persistent misconceptions — for example, students who still believe all plant cells always have chloroplasts, or who cannot articulate why the absence of a nucleus in a mature red blood cell is an advantage. These students are prioritised for support during the independent writing phase.
Observe Phase  VIDEO: Cell parts and their functions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12	The teacher now displays or distributes a structured three-column comparison card showing the three cells from the anchoring phenomenon: (1) mature red blood cell — biconcave disc, no nucleus, packed with haemoglobin; (2) white blood cell — irregular shape, large lobed nucleus, many lysosomes; (3) sukuma wiki leaf mesophyll cell — rectangular, cell wall, chloroplasts, large central vacuole. Each cell is shown as a diagram with five structural	Distribute or display the three-column comparison card. Give clear task instructions: 'You have 5 minutes individually. For each of the three cells on the card, fill in your table: one shared structure, one type-specific structure, one specialisation modification. Use evidence from your notes, your model, and the diagram in front of you.' Circulate during individual work. Do not answer content questions — redirect students to their notes: 'Check your organelle	The three-column comparison card re-anchors all abstract organelle learning back to the three specific cells from the driving phenomenon. This prevents the common failure mode where students can define organelles in isolation but cannot apply them to explain a specific cell type. The reconciliation step models scientific peer review at a classroom scale.	Teacher scans completed tables during circulation. Key diagnostic check: does the student correctly identify that the red blood cell lacks a nucleus as a specialisation modification (not simply as a deficiency), and can they link this to increased haemoglobin capacity? Students who describe the absence of a nucleus as damage or illness still carry a fundamental misconception and need a targeted one-sentence correction before the writing

 Search ARES for similar readings	features labelled. Students work individually for 5 minutes to complete a rapid identification task: for each cell, they must identify (a) one structure it SHARES with a generic cell, (b) one structure that is UNIQUE to its type (plant or animal), and (c) one structural modification that is linked directly to its specialised function. Students record their answers in their exercise books using a simple three-row table with the three column headings: Shared Structure, Type-Specific Structure, Specialisation Modification. After individual work, students compare answers with a neighbour and reconcile any differences.	function table from Lesson 3 or Lesson 4.' After 5 minutes, say: 'Compare your table with your neighbour. If you disagree on any cell, discuss for 2 minutes and decide together which answer is better supported by evidence.' After partner reconciliation, take whole-class responses for each cell using a show-of-hands consensus check: 'Hands up if you wrote cell membrane as the shared structure for red blood cells — good. Does anyone have a different shared structure? Tell me.' Explicitly validate that there is more than one correct answer for the shared structure column as long as the answer is scientifically accurate. WAIT TIME: after asking for a different answer, wait 6 seconds before confirming or correcting.		phase.
Explain Phase  VIDEO: Cell parts and their functions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	This is the core assessment activity of the lesson, lasting 15 minutes. Each student independently writes a full evidence-based explanation answering the driving question. Before writing begins, the teacher and class co-construct a four-part success criteria on the board. The four required parts are: (A) Name and explain at least two structures that ALL cells share and explain why these structures are essential to all living cells; (B) Name and explain at least two structures unique to plant cells and at least one structure unique to or more common in animal cells, with reference to the sukuma wiki leaf cell; (C) Explain how specialisation modifies or removes organelles to optimise function, using the red blood cell OR the white blood cell as your example; (D) Include one	Before writing begins, project or write the four-part success criteria on the board. Read each criterion aloud and ask: 'Any questions about what each part is asking?' Address clarification questions only — do not pre-answer the content. Then say: 'You have 15 minutes. Your notes, your diagrams, and your model are all available to you. This is not about memory — this is about whether you can use what you have learned to explain a real phenomenon. Begin.' Set a visible timer for 15 minutes. Circulate slowly and silently. Do not interrupt writers. If a student is stuck and raises their hand, lean in and say quietly: 'Read part A of the success criteria again and look at your comparison table from Lesson 4. What do ALL three cells on the card have in common?' Do	Allowing open notes transforms the writing task from a recall test into a genuine synthesis task — the cognitive demand is in organising and applying knowledge, not retrieving it. The four-part structure scaffolds the argument without dictating the content. Reading three student responses aloud after writing creates a whole-class norming moment that reveals the range of quality and helps all students self-assess.	Written explanations serve as the primary summative assessment for Sub-Strand 1.3. Teacher reads three responses aloud (with student permission and anonymised if preferred) — one strong, one developing, one partial — and asks the class: 'Which part of the success criteria does each response address well? Which part needs more evidence?' This models evaluative thinking and gives all students immediate feedback on quality standards before books are collected.

	Kenya-relevant example — this may be the sickle cell supporting phenomenon, the malaria parasite entering a red blood cell, or root hair absorption in bean seedlings from the school garden. Students write independently in silence. No notes are required to be closed — the teacher explicitly says this is a synthesis task, not a memory test, and students may use all their notes, diagrams, and models from the unit.	not provide content. With 3 minutes remaining, say: 'You have 3 minutes. If you have not reached part D, skip to it now — one Kenya-relevant example. You can fill in the middle after.' This prevents students from running out of time before demonstrating contextual application. After 15 minutes, say: 'Pens down. You are going to hear three student responses and I want you to listen for evidence — specific organelles, specific cells, specific functions.'		
Driving Question Board (DQB) Creation  VIDEO: Cell parts and their functions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	The class moves to the DQB display, which has been building since Lesson 1. In Lesson 7, students already sorted questions into answered, partially answered, and bridge categories. In this final lesson, students spend 4 minutes in a structured whole-class review. The teacher reads each remaining partially-answered or bridge question aloud. For each one, the class votes: has this lesson or their written explanation now answered it? If yes, a green tick sticker is added (or a green marker mark). If it remains a bridge question, a student volunteers to read aloud a one-sentence preview of where the answer will come from in Sub-Strand 1.4 Chemicals of Life. For example, the question 'What are cells actually made of at the chemical level?' receives the preview: 'In Sub-Strand 1.4 we will learn about carbohydrates, proteins, lipids, and nucleic acids — the molecules that build every organelle we have studied.' The DQB is then photographed by the teacher as a record of the unit inquiry arc.	Stand beside the DQB and say: 'This board has been our scientific memory for eight lessons. Every question you asked in Lesson 1 is still here. Let us find out which ones we have now answered — and which ones we are handing to our future selves.' Read each remaining question aloud. After each one, ask: 'Has our work today — or across this unit — given us a full answer to this question? Thumbs up for yes, thumbs sideways for partial, thumbs down for not yet.' Tally the class response and mark the card accordingly. For bridge questions, say: 'This question is not unanswered — it is waiting for more evidence. Who can give us one sentence about where that evidence will come from?' Cold-call a student. WAIT TIME: wait 5 seconds for the student to formulate the sentence. Do not rephrase or complete it for them. After the DQB review, say: 'I am going to photograph this board. This is your unit in question form — proof that you came in asking small questions and you are leaving asking bigger, better ones. That is what scientists do.'	The DQB closure ritual makes the inquiry arc of the entire unit visible and emotionally satisfying. Naming bridge questions as belonging to future science — rather than as failures or gaps — teaches students that unanswered questions are assets, not deficits. The thumbs vote gives every student a low-stakes opportunity to self-assess against the full unit.	The pattern of thumbs responses reveals whether the class as a whole has reached unit-level understanding or whether particular questions remain genuinely unresolved. Questions that receive predominantly thumbs-down responses even after eight lessons signal either a teaching gap or a question that was always beyond the unit scope — both are useful diagnostic information for teacher reflection.

Model Building Phase  VIDEO: Cell parts and their functions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Cell Transport (Flexbook) Source: CK-12  Search ARES for similar readings	In this final phase, lasting 5 minutes, the class co-constructs a consensus summary model on the board — not a 3D physical model (that was Lesson 7) but a conceptual map. The teacher draws three large overlapping circles on the board labelled: ALL CELLS, PLANT CELLS ONLY, ANIMAL CELLS ONLY. Students are called up one at a time to write one organelle or structural feature in the correct circle or overlap region. Each student must also speak one sentence justifying the placement. This continues until the board is populated with at least 10 to 12 features. After the map is complete, the teacher draws an arrow from the ALL CELLS circle and writes the label SPECIALISED CELLS — then asks: 'If a cell becomes specialised, which direction does it move — toward having more of these shared features, or fewer?' Students respond and the teacher draws connecting lines from specific specialised cells (red blood cell, guard cell, root hair cell) to show that specialisation always starts from the shared model and deviates from it. The lesson closes with the teacher previewing Sub-Strand 1.4.	Draw the three overlapping circles on the board clearly. Say: 'I am going to call you up one at a time. You write one feature in the correct region and say one sentence explaining why it belongs there. If the class agrees, it stays. If someone disagrees, we discuss for 30 seconds and vote.' Manage this briskly — each student should take no more than 20 to 25 seconds. After 10 to 12 features are placed, draw the specialised cells arrow and ask: 'Specialisation always starts from our generic model and then changes it. Can someone come up and draw a line from root hair cell to the feature that was modified to make it a root hair cell?' Allow one or two students to draw lines. Then close the lesson: 'We started this unit astonished by a blood smear. We now know why those cells look different — and we can explain it with evidence. But we still do not know what these organelles are actually built from at the chemical level. That is where we are going next: Sub-Strand 1.4, Chemicals of Life. The question shifts from what do cells look like to what are they made of and how does that chemistry make life possible.' Write the bridging question on the board: 'What are cells actually made of at the molecular level?' and leave it visible as students depart. WAIT TIME: after asking the specialisation arrow question, wait 6 seconds before calling a student forward.	The co-constructed consensus map externalises the class shared understanding in a form that every student contributes to and can see. Requiring a spoken justification for every placement ensures that surface participation (writing a word) is paired with conceptual accountability (explaining why). The closing preview of Sub-Strand 1.4 uses the same driving question format as Lesson 1, signalling to students that inquiry is continuous and that this lesson is an ending and a beginning simultaneously.	The consensus map reveals which organelles students can correctly place without prompting and which ones produce disagreement or incorrect placement. Persistent misplacements — for example, placing ribosomes in the PLANT CELLS ONLY circle — indicate that the ribosome-is-universal concept has not consolidated and should be addressed in the opening of Sub-Strand 1.4. The teacher photographs the completed map alongside the DQB as paired records of unit learning.
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D. TEACHER REFLECTION

After the lesson, the teacher should reflect on four questions. First: did students in their written explanations demonstrate genuine synthesis — connecting organelle

structure to specialised function to the Kenya-relevant example — or did they produce a list of definitions? Explanations that list organelle names without connecting them to the driving question indicate that the unit emphasised recall over reasoning and should be adjusted in future. Second: were there any organelles or concepts that were consistently absent or incorrect in the written explanations, suggesting insufficient instructional time or unclear explanation in earlier lessons? Third: did the DQB closure reveal any questions that the teacher had not anticipated, suggesting that student inquiry generated learning trajectories beyond the planned curriculum? These unanticipated questions are the most valuable data from the DQB and should be preserved for Sub-Strand 1.4 planning. Fourth: did the Kenya-relevant examples used by students show genuine contextual understanding — for example, explaining why sickle-shaped cells cannot pass through capillaries in the same way that biconcave cells can — or did students simply name a Kenya context without analysing it? Genuine contextual application is the hallmark of successful CBE implementation at this level.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We revisited the anchoring phenomenon — the blood smear showing red blood cells without nuclei next to white blood cells with large lobed nuclei, and the sukuma wiki leaf cell with its rectangular shape, chloroplasts, and large central vacuole. We observed a three-cell comparison card identifying shared structures, type-specific structures, and specialisation modifications for each cell. We observed the completed Driving Question Board from all eight lessons and reviewed which questions had been answered and which remain as bridges to Sub-Strand 1.4.
What did I learn?	All cells share core structures including a cell membrane, cytoplasm, ribosomes, and DNA, because these are essential to the basic processes of life including protein synthesis, energy use, and reproduction. Plant cells are distinguished by a cell wall made of cellulose, chloroplasts for photosynthesis, and a large central vacuole for turgor pressure — all visible in the sukuma wiki leaf. Animal cells may possess centrioles for cell division. Specialised cells are not defective generic cells — they are optimised deviations: the red blood cell loses its nucleus and gains haemoglobin capacity; the guard cell changes shape to control the stomatal opening in the sukuma wiki leaf; the root hair cell extends to maximise absorption surface area. Sickle cell disease demonstrates that even a small structural change — caused by a single gene mutation — completely disrupts a specialised cell function and has significant health consequences in Kenyan communities where the sickle cell trait is present.
How does this explain the phenomenon?	We can now answer the driving question with evidence: the cells in our body and in the sukuma wiki look different from each other because specialisation requires each cell to prioritise certain functions over others, and the most efficient way to do that is to modify or remove structures that are not needed while amplifying or adding structures that are. The red blood cell without a nucleus is not missing something — it has made space for more haemoglobin so it can carry more oxygen. The sukuma wiki leaf cell is not more complex than a cheek cell — it has additional structures because it must harvest light energy from the sun, something an animal cell never needs to do. All of these cells — red blood cell, white blood cell, leaf cell, root hair cell — are still cells because they all share the fundamental architecture of membrane, cytoplasm, and genetic material. The difference is in what has been added, removed, or modified in service of a specific job in a specific organism living in a specific environment.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.